

# Hadoop Cluster Administration

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# **Cluster & Tools Installation**

# Resources Used

16 GB Ram

Task Manager

File Options View

Processes Performance App history Startup Users Details Services

| Name                                   | Status | 13%<br>CPU | 74%<br>Memory | 2%<br>Disk | 6%<br>Network |
|----------------------------------------|--------|------------|---------------|------------|---------------|
| Apps (10)                              |        |            |               |            |               |
| > Bitvise SSH Client - xterm termin... |        | 0%         | 6,5 MB        | 0 MB/s     | 0 Mbps        |
| > Bitvise SSH Client (32 bit)          |        | 0%         | 4,4 MB        | 0 MB/s     | 0 Mbps        |
| > Google Chrome (26)                   |        | 0,7%       | 489,9 MB      | 0 MB/s     | 4,3 Mbps      |
| > Microsoft PowerPoint (2)             |        | 0%         | 89,3 MB       | 0 MB/s     | 0 Mbps        |
| > Spotify (10)                         |        | 0,9%       | 346,7 MB      | 0,1 MB/s   | 0 Mbps        |
| > Task Manager                         |        | 0,7%       | 25,8 MB       | 0,1 MB/s   | 0 Mbps        |
| > VirtualBox Manager                   |        | 0,1%       | 74,2 MB       | 0 MB/s     | 0 Mbps        |
| > VirtualBox Virtual Machine           |        | 7,7%       | 2.348,4 MB    | 0,1 MB/s   | 0 Mbps        |
| > VirtualBox Virtual Machine           |        | 0,3%       | 2.285,6 MB    | 0 MB/s     | 0 Mbps        |
| > VirtualBox Virtual Machine           |        | 0%         | 2.342,0 MB    | 0 MB/s     | 0 Mbps        |
| Background processes (84)              |        |            |               |            |               |
| AqqregatorHost.exe                     |        | 0%         | 0,5 MB        | 0 MB/s     | 0 Mbps        |

<

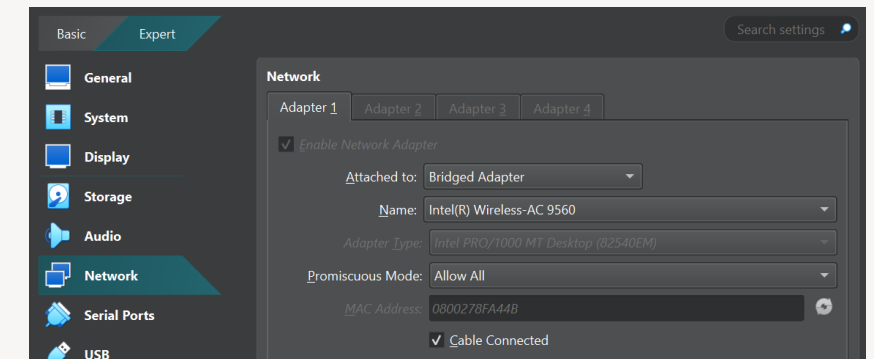
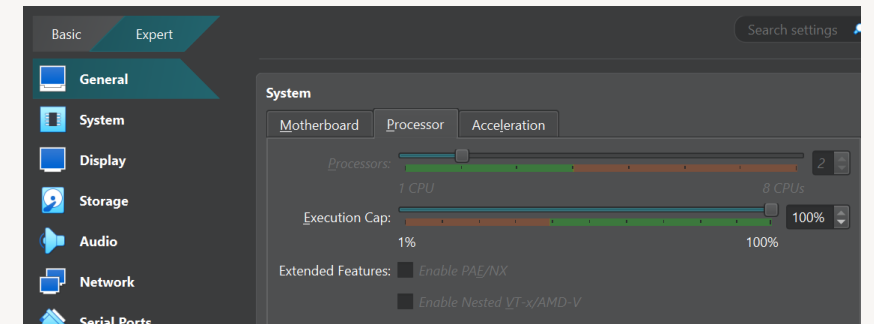
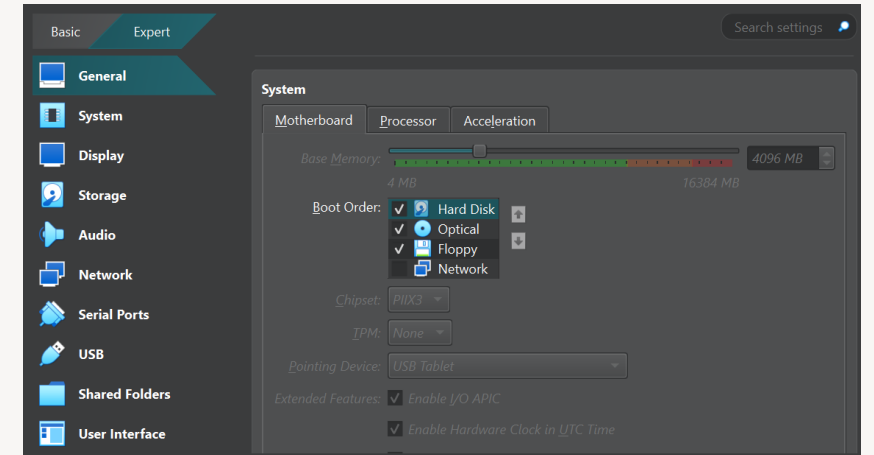
>

^ Fewer details

End task

# Nodes

Set all nodes configuration to the following screenshots, all nodes should be granted similar resources if possible.

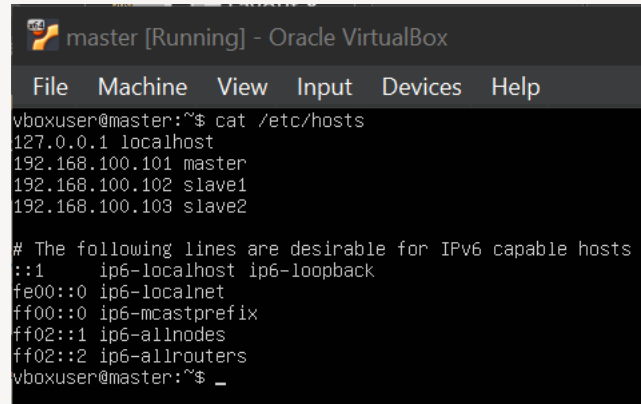


# Prerequisites (Install on all 3 nodes)

```
sudo apt-get update
sudo apt-get upgrade
sudo apt-get install net-tools
sudo apt-get install nano
sudo apt-get install openjdk-11-jdk
sudo apt-get install ssh
```

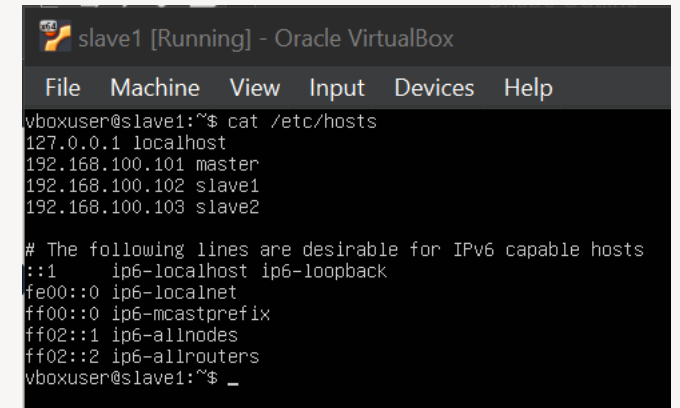
```
sudo apt-get install maven
sudo apt-get install build-essential cmake pkg-config zlib1g-dev libsnappy-dev
mvn clean package -Pnative -DskipTests
```

```
sudo addgroup hadoop
sudo adduser -ingroup hadoop hdadmin
sudo adduser hdadmin sudo
su hdadmin
ssh-keygen -t rsa -P ""
cat $HOME/.ssh/id_rsa.pub >> $HOME/.ssh/authorized_keys
chmod 600 $HOME/.ssh/authorized_keys
```



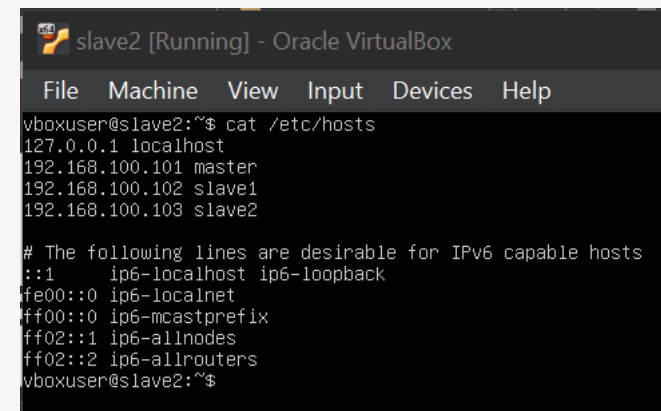
```
master [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@master:~$ cat /etc/hosts
127.0.0.1 localhost
192.168.100.101 master
192.168.100.102 slave1
192.168.100.103 slave2

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
vboxuser@master:~$ _
```



```
slave1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@slave1:~$ cat /etc/hosts
127.0.0.1 localhost
192.168.100.101 master
192.168.100.102 slave1
192.168.100.103 slave2

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
vboxuser@slave1:~$ _
```



```
slave2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@slave2:~$ cat /etc/hosts
127.0.0.1 localhost
192.168.100.101 master
192.168.100.102 slave1
192.168.100.103 slave2

# The following lines are desirable for IPv6 capable hosts
::1          ip6-localhost ip6-loopback
fe00::0      ip6-localnet
ff00::0      ip6-mcastprefix
ff02::1      ip6-allnodes
ff02::2      ip6-allrouters
vboxuser@slave2:~$
```

# Network Connection

Set Node Network Interface to “Bridged”, promiscuous mode “Allow All”

```
ip addr
ping <all nodes>
Ifconfig enp0s3 down/up
```

```
master [Running] - Oracle VirtualBox
File Machine View Input Devices Help
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9f:a4:4b brd ff:ff:ff:ff:ff:ff
    inet 192.168.70.173/24 metric 100 brd 192.168.70.255 scope global dynamic enp0s3
        valid_lft 3568sec preferred_lft 3568sec
    inet6 fe80::a00:27ff:fe8f:a44b/64 scope link
        valid_lft forever preferred_lft forever
hdadmin@master:/home/vboxuser$ ping 192.168.70.201
PING 192.168.70.201 (192.168.70.201) 56(84) bytes of data.
64 bytes from 192.168.70.201: icmp_seq=1 ttl=64 time=2.26 ms
64 bytes from 192.168.70.201: icmp_seq=2 ttl=64 time=1.98 ms
64 bytes from 192.168.70.201: icmp_seq=3 ttl=64 time=1.11 ms
64 bytes from 192.168.70.201: icmp_seq=4 ttl=64 time=0.930 ms
64 bytes from 192.168.70.201: icmp_seq=5 ttl=64 time=1.68 ms
64 bytes from 192.168.70.201: icmp_seq=6 ttl=64 time=1.59 ms
64 bytes from 192.168.70.201: icmp_seq=7 ttl=64 time=1.23 ms
^C
--- 192.168.70.201 ping statistics ---
7 packets transmitted, 7 received, 0% packet loss, time 6094ms
rtt min/avg/max/mdev = 0.930/1.539/2.255/0.444 ms
hdadmin@master:/home/vboxuser$ ping 192.168.70.15
PING 192.168.70.15 (192.168.70.15) 56(84) bytes of data.
64 bytes from 192.168.70.15: icmp_seq=1 ttl=64 time=2.41 ms
64 bytes from 192.168.70.15: icmp_seq=2 ttl=64 time=1.69 ms
64 bytes from 192.168.70.15: icmp_seq=3 ttl=64 time=1.37 ms
64 bytes from 192.168.70.15: icmp_seq=4 ttl=64 time=1.46 ms
64 bytes from 192.168.70.15: icmp_seq=5 ttl=64 time=2.30 ms
^C
--- 192.168.70.15 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4047ms
rtt min/avg/max/mdev = 1.372/1.846/2.408/0.428 ms
hdadmin@master:/home/vboxuser$
```

```
slave1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
hdadmin@slave1:/home/vboxuser$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:33:6a:fd brd ff:ff:ff:ff:ff:ff
    inet 192.168.70.201/24 metric 100 brd 192.168.70.255 scope global dynamic enp0s3
        valid_lft 3554sec preferred_lft 3554sec
    inet6 fe80::a00:27ff:fe33:6afd/64 scope link
        valid_lft forever preferred_lft forever
hdadmin@slave1:/home/vboxuser$ ping 192.168.70.173
PING 192.168.70.173 (192.168.70.173) 56(84) bytes of data.
64 bytes from 192.168.70.173: icmp_seq=1 ttl=64 time=0.959 ms
64 bytes from 192.168.70.173: icmp_seq=2 ttl=64 time=1.41 ms
64 bytes from 192.168.70.173: icmp_seq=3 ttl=64 time=1.22 ms
^C
--- 192.168.70.173 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2003ms
rtt min/avg/max/mdev = 0.959/1.196/1.408/0.184 ms
hdadmin@slave1:/home/vboxuser$ ping 192.168.70.15
PING 192.168.70.15 (192.168.70.15) 56(84) bytes of data.
64 bytes from 192.168.70.15: icmp_seq=1 ttl=64 time=1.72 ms
64 bytes from 192.168.70.15: icmp_seq=2 ttl=64 time=2.26 ms
64 bytes from 192.168.70.15: icmp_seq=3 ttl=64 time=1.68 ms
^C
--- 192.168.70.15 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.682/1.885/2.260/0.265 ms
hdadmin@slave1:/home/vboxuser$
```

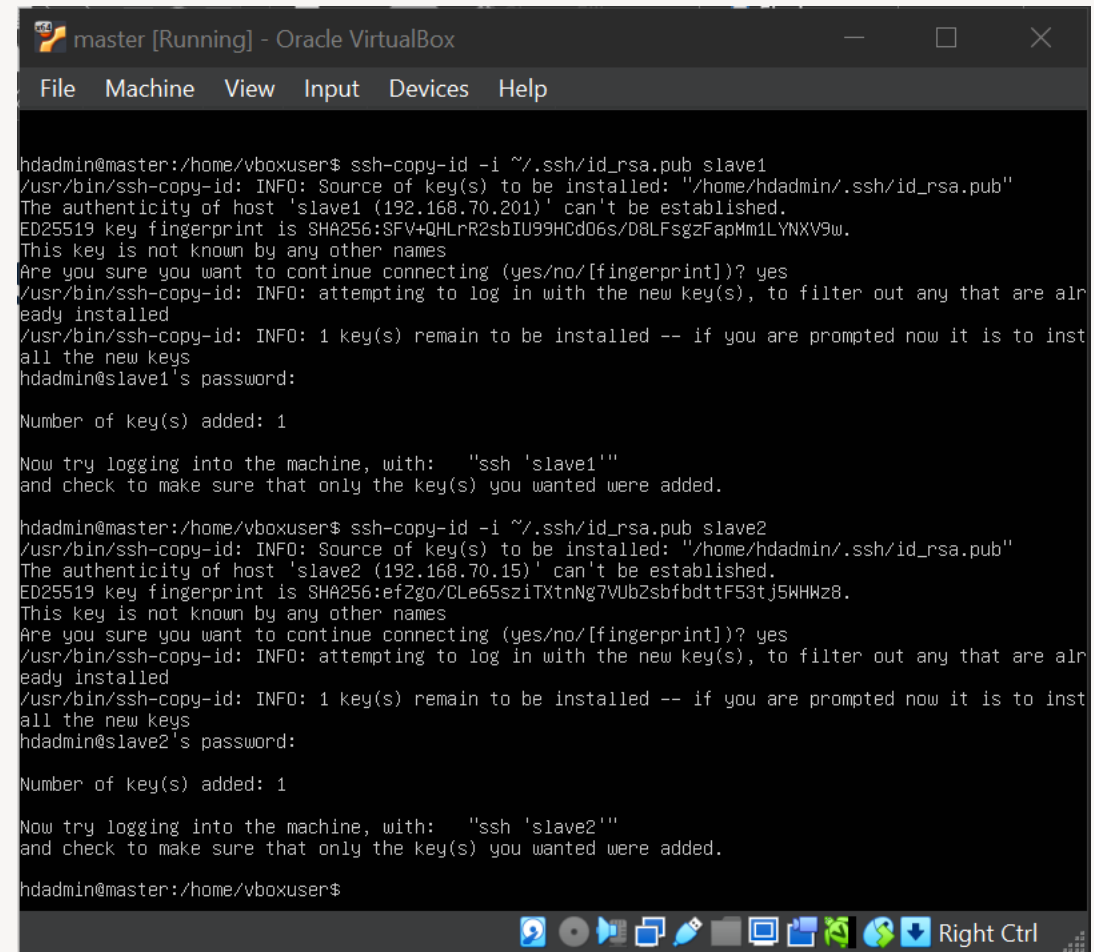
```
slave2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
hdadmin@slave2:/home/vboxuser$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:be:04:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.70.15/24 metric 100 brd 192.168.70.255 scope global dynamic enp0s3
        valid_lft 3570sec preferred_lft 3570sec
    inet6 fe80::a00:27ff:febe:41e/64 scope link
        valid_lft forever preferred_lft forever
hdadmin@slave2:/home/vboxuser$ ping 192.168.70.173
PING 192.168.70.173 (192.168.70.173) 56(84) bytes of data.
64 bytes from 192.168.70.173: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.70.173: icmp_seq=2 ttl=64 time=1.78 ms
64 bytes from 192.168.70.173: icmp_seq=3 ttl=64 time=1.03 ms
^C
--- 192.168.70.173 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 1.033/1.291/1.776/0.342 ms
hdadmin@slave2:/home/vboxuser$ ping 192.168.70.201
PING 192.168.70.201 (192.168.70.201) 56(84) bytes of data.
64 bytes from 192.168.70.201: icmp_seq=1 ttl=64 time=1.22 ms
64 bytes from 192.168.70.201: icmp_seq=2 ttl=64 time=1.53 ms
64 bytes from 192.168.70.201: icmp_seq=3 ttl=64 time=2.05 ms
^C
--- 192.168.70.201 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 1.224/1.623/2.053/0.339 ms
hdadmin@slave2:/home/vboxuser$
```

# Change the IP Address in /etc/hosts (3 Nodes)

[Master]

```
ssh-copy-id -i ~/.ssh/id_rsa.pub slave1  
ssh-copy-id -i ~/.ssh/id_rsa.pub slave2
```

```
GNU nano 6.2  
127.0.0.1 localhost  
192.168.70.173 master  
192.168.70.201 slave1  
192.168.70.15 slave2
```



```
master [Running] - Oracle VirtualBox  
File Machine View Input Devices Help  
  
hdadmin@master:/home/vboxuser$ ssh-copy-id -i ~/.ssh/id_rsa.pub slave1  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hdadmin/.ssh/id_rsa.pub"  
The authenticity of host 'slave1 (192.168.70.201)' can't be established.  
ED25519 key fingerprint is SHA256:SFV+QHLrR2sbIU99HCd06s/D8LFsgZFapMm1LYNXV9w.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are al  
eady installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to inst  
all the new keys  
hdadmin@slave1's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh 'slave1'"  
and check to make sure that only the key(s) you wanted were added.  
  
hdadmin@master:/home/vboxuser$ ssh-copy-id -i ~/.ssh/id_rsa.pub slave2  
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/hdadmin/.ssh/id_rsa.pub"  
The authenticity of host 'slave2 (192.168.70.15)' can't be established.  
ED25519 key fingerprint is SHA256:efZgo/CLe65sziTXtnNg7VUb2sbfbdtF53tj5WHWz8.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are al  
eady installed  
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to inst  
all the new keys  
hdadmin@slave2's password:  
  
Number of key(s) added: 1  
  
Now try logging into the machine, with: "ssh 'slave2'"  
and check to make sure that only the key(s) you wanted were added.  
  
hdadmin@master:/home/vboxuser$
```

# Change the IP Address in /etc/hosts (3 Nodes)

Now you can connect through ssh via bitvise / putty.

[Master SSH]

ssh slave1

ssh slave2

```
HDP.tlp - hdadmin@192.168.70.173:22 - Bitvise xterm - hdadmin@master: ~
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 5.15.0-126-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Nov 30 10:11:33 AM UTC 2024

System load:  0.0          Processes:           116
Usage of /:   6.0% of 62.69GB Users logged in:       0
Memory usage: 8%          IPv4 address for enp0s3: 192.168.70.173
Swap usage:   0%

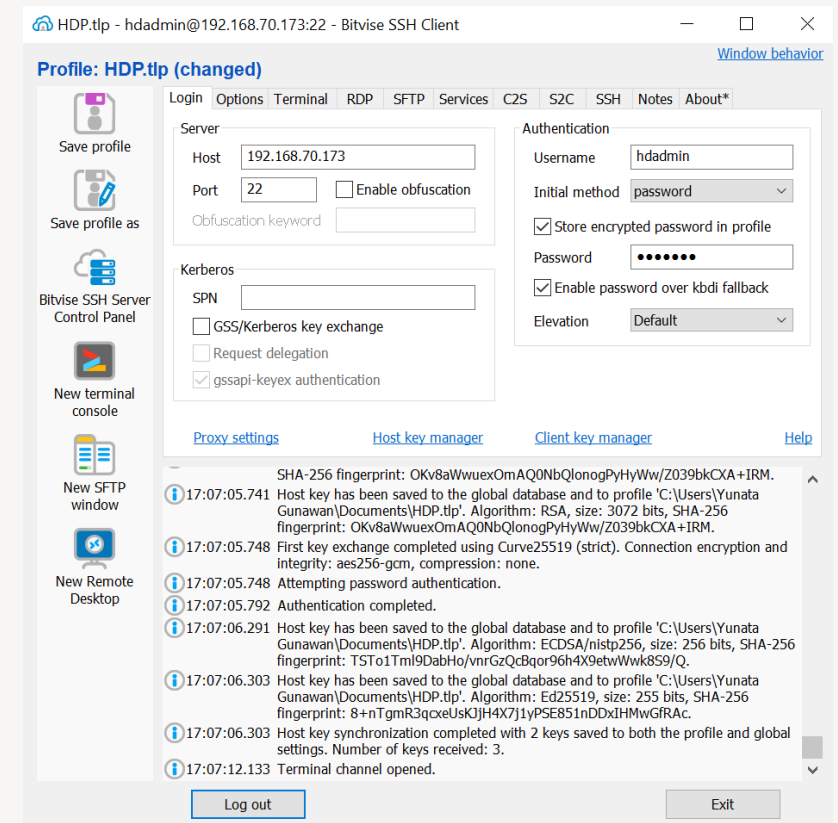
Expanded Security Maintenance for Applications is not enabled.

5 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Sat Nov 30 10:07:12 2024 from 192.168.70.29
hdadmin@master:~$
```



# Hadoop Configuration #1

## [Master Node]

```
cd /usr/local/  
wget https://dlcdn.apache.org/hadoop/common/hadoop-3.4.1/hadoop-3.4.1.tar.gz  
sudo tar xvzf hadoop-3.4.1.tar.gz  
sudo mv hadoop-3.4.1 hadoop  
sudo nano ~/.bashrc
```

## [nano]

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64  
export HADOOP_HOME=/usr/local/hadoop  
export PATH=$PATH:$HADOOP_HOME/bin  
export PATH=$PATH:$HADOOP_HOME/sbin  
export HADOOP_MAPRED_HOME=$HADOOP_HOME  
export HADOOP_COMMON_HOME=$HADOOP_HOME  
export HADOOP_HDFS_HOME=$HADOOP_HOME  
export YARN_HOME=$HADOOP_HOME  
export HADOOP_COMMON_LIB_NATIVE_DIR=$HADOOP_HOME/lib/native  
export HADOOP_OPTS="-Djava.library.path=$HADOOP_HOME/lib"  
export HADOOP_CONF_DIR=/usr/local/hadoop/etc/hadoop  
export HADOOP_CLASSPATH=${JAVA_HOME}/lib/tools.jar  
export HDFS_NAMENODE_USER=hdadmin  
export HDFS_DATANODE_USER=hdadmin  
export HDFS_SECONDARYNAMENODE_USER=hdadmin
```

## [Master Node]

```
source ~/.bashrc
```

# Hadoop Configuration #2

```
sudo nano $HADOOP_INSTALL/etc/hadoop/hadoop-env.sh
```

[Nano]

```
export JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64
```

[Master, Slave1, Slave2]

Use IP Address instead of hostname

# core-site.xml

```
<!-- The name of the default filesystem (usually Namenode, port 8020) -->
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://master:8020</value>
</property>

<!-- Hadoop Temporary Directory -->
<property>
  <name>hadoop.tmp.dir</name>
  <value>/app/hadoop/tmp</value>
  <description>Temporary directory for Hadoop's use. It will be created if it doesn't exist.<>
</property>

<!-- Local directory for storage of system logs -->
<property>
  <name>hadoop.log.dir</name>
  <value>/app/hadoop/log</value>
</property>

<!-- Cluster Security Setting -->
<property>
  <name>hadoop.security.authentication</name>
  <value>simple</value>
</property>
```

# hdfs-site.xml (General)

```
<!-- Replication factor for data storage -->
<property>
  <name>dfs.replication</name>
  <value>3</value>
  <description>Number of replicas of each block in the HDFS cluster</description>
</property>

<property>
  <name>dfs.permissions</name>
  <value>true</value>
  <description>Enable permission checks in HDFS</description>
</property>

<!-- Directory for NameNode to store its metadata -->
<property>
  <name>dfs.namenode.name.dir</name>
  <value>file:///usr/local/hadoop_store/hdfs/namenode</value>
  <description>Directory for storing NameNode metadata (fsimage and edits)</description>
</property>

<!-- Directory for DataNode to store data blocks -->
<property>
  <name>dfs.datanode.data.dir</name>
  <value>file:///usr/local/hadoop_store/hdfs/datanode</value>
  <description>Directory for storing DataNode data blocks</description>
</property>

<!-- DataNode block size -->
<property>
  <name>dfs.blocksize</name>
  <value>134217728</value> <!-- 128 MB -->
</property>

<property>
  <name>dfs.namenode.secondary.http-address</name>
  <value>0.0.0.0:50090</value>
</property>
```

# hdfs-site.xml (Namenode)

```
<!-- Directory for NameNode to store its metadata -->  
<property>  
  <name>dfs.namenode.name.dir</name>  
  <value>file:///usr/local/hadoop_store/hdfs/namenode</value>  
  <description>Directory for storing NameNode metadata (fsimage and edits)</description>  
</property>
```

# hdfs-site.xml (Datanode)

```
<!-- Directory for DataNode to store data blocks -->  
<property>  
  <name>dfs.datanode.data.dir</name>  
  <value>file:///usr/local/hadoop_store/hdfs/datanode</value>  
  <description>Directory for storing DataNode data blocks</description>  
</property>
```

# hdfs-site.xml (Secondary Namenode)

```
<!-- Directory for Secondary Namenode -->
<property>
  <name>dfs.namenode.name.dir</name>
  <value>file:///usr/local/hadoop_store/hdfs/secondarynamenode</value>
  <description>Directory where Namenode checkpoint image files are stored.</description>
</property>

<!-- The frequency (in seconds) for creating checkpoints -->
<property>
  <name>dfs.namenode.checkpoint.period</name>
  <value>3600</value>
  <description>Period (in seconds) between each checkpoint creation.</description>
</property>

<!-- The number of edit log segments to merge during a checkpoint -->
<property>
  <name>dfs.namenode.checkpoint.txns</name>
  <value>100000</value>
  <description>The number of transactions after which a checkpoint will be triggered.</descri>
</property>

<!-- Enable or disable the Checkpoint Node (optional) -->
<property>
  <name>dfs.namenode.checkpoint.enabled</name>
  <value>true</value>
  <description>Whether checkpointing should be enabled for the Secondary NameNode.</descripti>
</property>
```

# mapred-site.xml

```
<property>  
  <name>mapreduce.framework.name</name>  
  <value>yarn</value>  
</property>
```

```
<property>  
  <name>mapreduce.application.classpath</name>  
  
<value>${HADOOP_MAPRED_HOME}/share/hadoop/mapreduce/*:${HADOOP_MAPRED_HOME}/share/hadoop/map  
red>  
</property>
```

```
<!-- The address of the Jobtracker for MapReduce -->  
<property>  
  <name>mapreduce.jobtracker.address</name>  
  <value>master:54311</value>  
</property>
```

# yarn-site.xml

```
<property>
  <name>yarn.nodemanager.aux-services</name>
  <value>mapreduce_shuffle</value>
</property>

<property>
  <name>yarn.nodemanager.env-whitelist</name>
  <value>JAVA_HOME,HADOOP_COMMON_HOME,HADOOP_HDFS_HOME,HADOOP_CONF_DIR,CLASSPATH_PREPEND_DIST>
</property>

<property>
  <name>yarn.resourcemanager.scheduler.address</name>
  <value>master:8030</value>
</property>

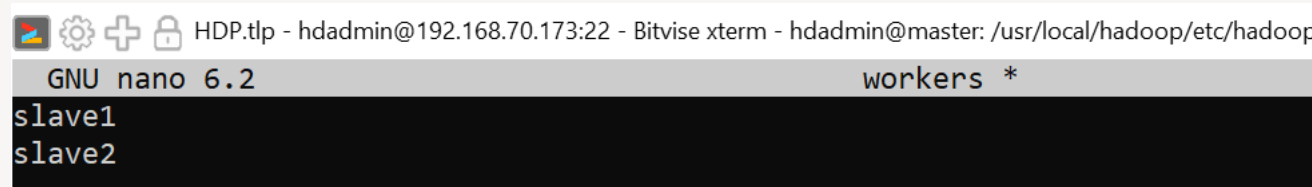
<property>
  <name>yarn.resourcemanager.address</name>
  <value>master:8032</value>
</property>

<property>
  <name>yarn.resourcemanager.webapp.address</name>
  <value>master:8088</value>
</property>

<property>
  <name>yarn.resourcemanager.resource-tracker.address</name>
  <value>master:8031</value>
</property>

<property>
  <name>yarn.resourcemanager.admin.address</name>
  <value>master:8033</value>
</property>
```

# Worker Config (All Nodes)



```
HDP.tlp - hdadmin@192.168.70.173:22 - Bitvise xterm - hdadmin@master: /usr/local/hadoop/etc/hadoop
GNU nano 6.2 workers *
slave1
slave2
```

[Master]

```
sudo scp -r /usr/local/hadoop/ hdadmin@slave1:~/hadoop/
sudo scp -r /usr/local/hadoop/ hdadmin@slave2:~/hadoop/
```

[Slave1, Slave2]

```
sudo mv ~/hadoop/ /usr/local/hadoop/
```

[Master, Slave1, Slave2]

```
sudo ./hdfs namenode -format
sudo chown -R hdadmin:hadoop /usr/local/hadoop_store
sudo chown -R hdadmin:hadoop /usr/local/hadoop
sudo chown -R hdadmin:hadoop /app
```

*\*to make master working as a datanode too*

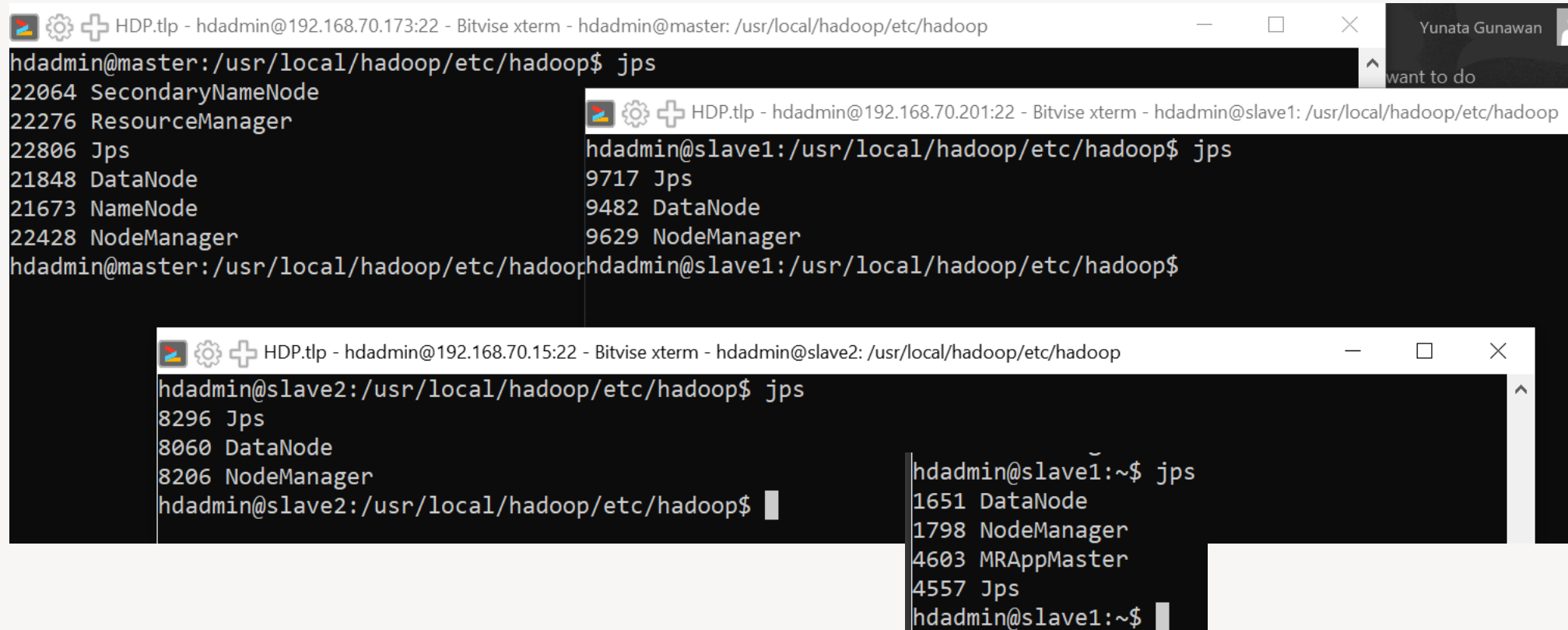


```
GNU nano 6.2 workers
master
slave1
slave2
```

## **Note:**

- ) Make sure the folder path of configuration exists and user have permission to sudo**
- ) Configuration files is same on clusters, difference who is namenode, datanode only located in “workers” file (all nodes)**
- ) Make sure the cluster in same network**

# Result (Hadoop 3 Nodes)



The image displays three terminal windows from a Bitvise xterm application, showing the output of the 'jps' command on different Hadoop nodes. The windows are titled 'HDP.tlp - hdadmin@192.168.70.173:22 - Bitvise xterm - hdadmin@master: /usr/local/hadoop/etc/hadoop', 'HDP.tlp - hdadmin@192.168.70.201:22 - Bitvise xterm - hdadmin@slave1: /usr/local/hadoop/etc/hadoop', and 'HDP.tlp - hdadmin@192.168.70.15:22 - Bitvise xterm - hdadmin@slave2: /usr/local/hadoop/etc/hadoop'. The master node shows processes for SecondaryNameNode, ResourceManager, Jps, DataNode, NameNode, and NodeManager. Slave1 shows Jps, DataNode, and NodeManager. Slave2 shows Jps, DataNode, and NodeManager. A fourth terminal window is partially visible on the right, showing Jps, DataNode, NodeManager, MRAppMaster, and Jps on slave1.

```
HDP.tlp - hdadmin@192.168.70.173:22 - Bitvise xterm - hdadmin@master: /usr/local/hadoop/etc/hadoop
hdadmin@master:/usr/local/hadoop/etc/hadoop$ jps
22064 SecondaryNameNode
22276 ResourceManager
22806 Jps
21848 DataNode
21673 NameNode
22428 NodeManager
hdadmin@master:/usr/local/hadoop/etc/hadoop$

HDP.tlp - hdadmin@192.168.70.201:22 - Bitvise xterm - hdadmin@slave1: /usr/local/hadoop/etc/hadoop
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ jps
9717 Jps
9482 DataNode
9629 NodeManager
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$

HDP.tlp - hdadmin@192.168.70.15:22 - Bitvise xterm - hdadmin@slave2: /usr/local/hadoop/etc/hadoop
hdadmin@slave2:/usr/local/hadoop/etc/hadoop$ jps
8296 Jps
8060 DataNode
8206 NodeManager
hdadmin@slave2:/usr/local/hadoop/etc/hadoop$

HDP.tlp - hdadmin@192.168.70.201:22 - Bitvise xterm - hdadmin@slave1: /usr/local/hadoop/etc/hadoop
hdadmin@slave1:~$ jps
1651 DataNode
1798 NodeManager
4603 MRAppMaster
4557 Jps
hdadmin@slave1:~$
```

# Resource Manager Web UI

<Namenode / SecondaryNamenode IP Address>:8088/cluster



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NEW\_SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

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Scheduler

Tools

About the Cluster

Cluster Metrics

| Apps Submitted | Apps Pending | Apps Running | Apps Completed | Containers Running | Used Resources         | Total Resources         | Reserved Resources     | Physical Mem Used % | Physical VCores Used % |
|----------------|--------------|--------------|----------------|--------------------|------------------------|-------------------------|------------------------|---------------------|------------------------|
| 1              | 0            | 0            | 1              | 0                  | <memory:0 B, vCores:0> | <memory:8 GB, vCores:2> | <memory:0 B, vCores:0> | 19                  | 0                      |

Cluster Nodes Metrics

| Active Nodes | Decommissioning Nodes | Decommissioned Nodes | Lost Nodes | Unhealthy Nodes | Rebooted Nodes | Shutdown Nodes |
|--------------|-----------------------|----------------------|------------|-----------------|----------------|----------------|
| 2            | 0                     | 0                    | 0          | 0               | 0              | 0              |

Scheduler Metrics

| Scheduler Type     | Scheduling Resource Type     | Minimum Allocation     | Maximum Allocation      | Maximum Cluster Application Priority | Scheduler Busy % | RM Dispatcher EventQueue Size | Scheduler Dispatcher EventQueue Size |
|--------------------|------------------------------|------------------------|-------------------------|--------------------------------------|------------------|-------------------------------|--------------------------------------|
| Capacity Scheduler | [memory-mb (unit=M), vcores] | <memory:256, vCores:1> | <memory:4096, vCores:1> | 0                                    | 0                | 0                             | 0                                    |

Cluster overview

Cluster ID:

1733067204611

ResourceManager state:

STARTED

ResourceManager HA state:

active

ResourceManager HA zookeeper connection state:

Could not find leader elector. Verify both HA and automatic failover are enabled.

ResourceManager RMStateStore:

org.apache.hadoop.yarn.server.resourcemanager.recovery.NullRMStateStore

ResourceManager started on:

Sun Dec 01 15:33:24 +0000 2024

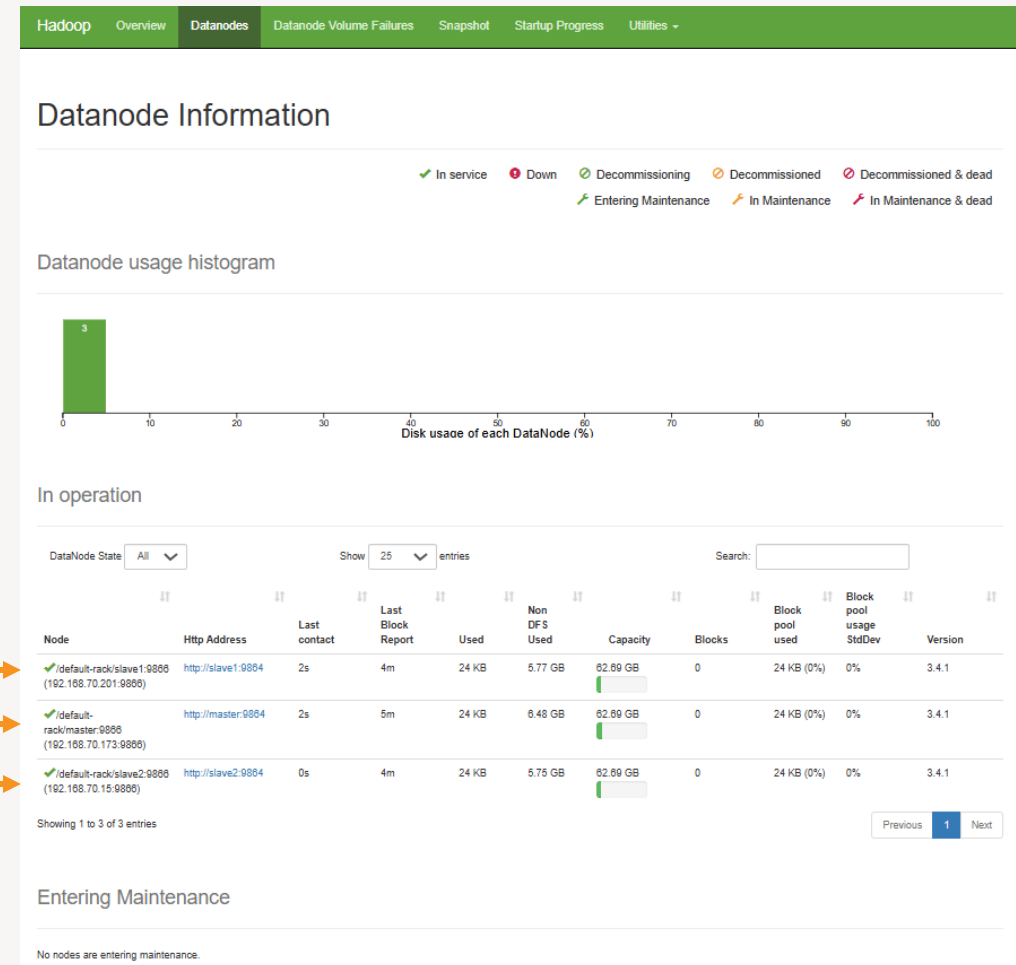
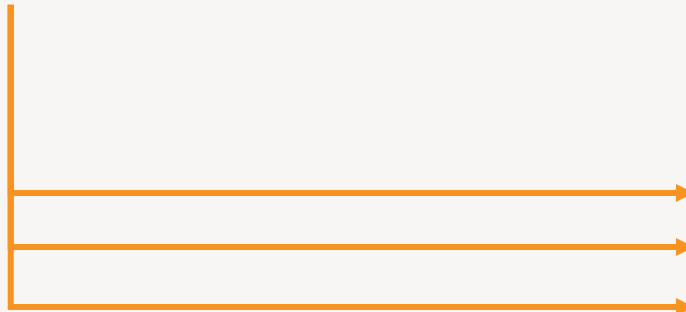
ResourceManager version:

3.4.1 from 4d7825309348956336b8f06a08322b78422849b1 by mthakur source checksum 934da0c5743762b7851cfcff9f8ca2 on 2024-10-09T15:14Z

# NameNode Web Interface

<Namenode IP Address>:9870/

**There is 3 node in here**



# Accessing HDFS Through Namenode

[HDFS]

-) `hadoop fs -ls <RELATIVE_FOLDER_PATH>`

-) `hadoop fs -mkdir <FOLDER_NAME>`

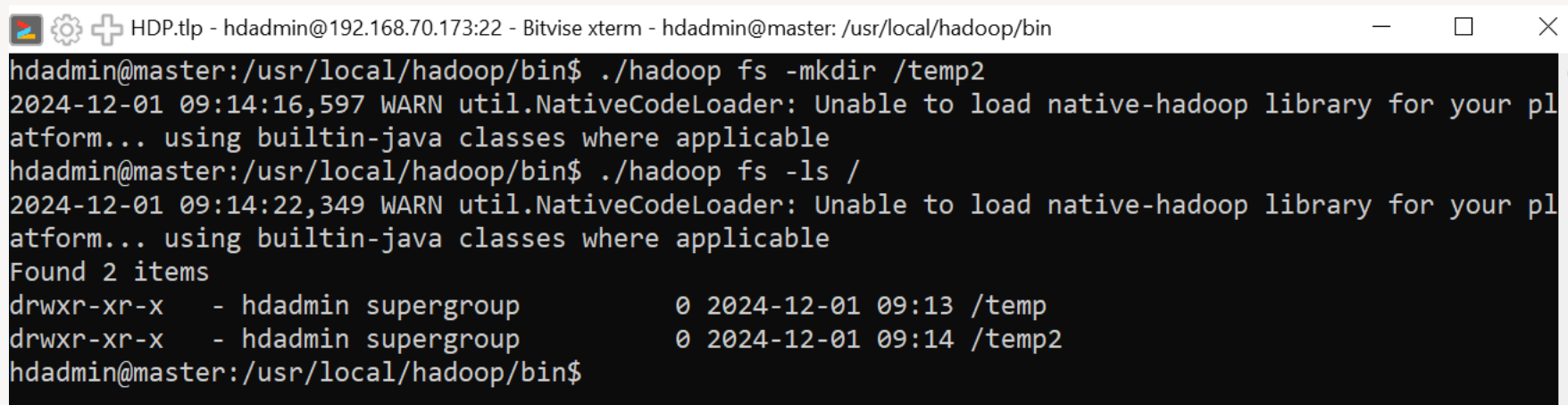
-) `hadoop fs -copyFromLocal <LINUX_RELATIVE_FILEPATH> <HDFS_RELATIVE_FILEPATH>`

-) `hadoop fs -cp <RELATIVE_PATH> <TO_RELATIVE_PATH>` | example: `hadoop fs -cp ./<FOLDER_NAME>/data.txt ./<FOLDER_NAME>/data2.txt`

-) `hadoop fs -rm <RELATIVE_FILENAME>` | example: `hadoop fs -rm ./<FOLDER_NAME>/*`

-) `hadoop fs -rmdir <RELATIVE_DIR>`

-) `hadoop fs`



```
HDP.tlp - hdadmin@192.168.70.173:22 - Bitwise xterm - hdadmin@master: /usr/local/hadoop/bin
hdadmin@master:/usr/local/hadoop/bin$ ./hadoop fs -mkdir /temp2
2024-12-01 09:14:16,597 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
hdadmin@master:/usr/local/hadoop/bin$ ./hadoop fs -ls /
2024-12-01 09:14:22,349 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Found 2 items
drwxr-xr-x  - hdadmin supergroup          0 2024-12-01 09:13 /temp
drwxr-xr-x  - hdadmin supergroup          0 2024-12-01 09:14 /temp2
hdadmin@master:/usr/local/hadoop/bin$
```

# WordCount.Java

nano WordCount.java

Copy code from here:

<https://hadoop.apache.org/docs/stable/hadoop-mapreduce-client/hadoop-mapreduce-client-core/MapReduceTutorial.html>

[Master]

Compile -> `hadoop com.sun.tools.javac.Main WordCount.java`

Create Jar -> `jar cf wc.jar WordCount*.class`

`hadoop fs -mkdir /user/joe/wordcount/input`

`nano file01`

`nano file02`

`hadoop fs -copyFromLocal ~/source_code/file01 /user/joe/wordcount/input/file01`

`hadoop fs -copyFromLocal ~/source_code/file02 /user/joe/wordcount/input/file02`

`hadoop jar wc.jar WordCount /user/joe/wordcount/input /user/joe/wordcount/output`

# Pending State (Resources + Permission)

← → ↻ ⚠ Not secure 192.168.70.173:8088/cluster/apps/ACCEPTED



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► Tools

Cluster Metrics

| Apps Submitted | Apps Pending | Apps Running | Apps Pending |
|----------------|--------------|--------------|--------------|
| 2              | 2            | 0            | 0            |

Cluster Nodes Metrics

| Active Nodes | Decommissioning Nodes |
|--------------|-----------------------|
| 0            | 0                     |

Scheduler Metrics

| Scheduler Type     | Scheduling Resource Type                               |
|--------------------|--------------------------------------------------------|
| Capacity Scheduler | [memory-mb (unit=M), vcores] <memory:1024MB, vcores:1> |

Show 20 ▼ entries

| ID                                             | User    | Name       | Application Type | Application Tags | Queue |
|------------------------------------------------|---------|------------|------------------|------------------|-------|
| <a href="#">application_1733046590212_0002</a> | hdadmin | word count | MAPREDUCE        |                  | root  |
| <a href="#">application_1733046590212_0001</a> | hdadmin | word count | MAPREDUCE        |                  | root  |

Showing 1 to 2 of 2 entries

# Fixing Resource Allocation

[yarn-site.xml]

```
<property>
  <description>Max available memory on each data node.</description>
  <name>yarn.nodemanager.resource.memory-mb</name>
  <value>4096</value>
</property>

<property>
  <description>Max available cores data node.</description>
  <name>yarn.nodemanager.resource.cpu-vcores</name>
  <value>2</value>
</property>

<property>
  <description>Minimum allocation unit.</description>
  <name>yarn.scheduler.minimum-allocation-mb</name>
  <value>256</value>
</property>

<property>
  <description>Max allocation unit.</description>
  <name>yarn.scheduler.maximum-allocation-mb</name>
  <value>4096</value>
</property>

<property>
  <description>Minimum increment setting - set to same as min-allocation</description>
  <name>yarn.scheduler.increment-allocation-mb</name>
  <value>256</value>
</property>
```

[mapred-site.xml]

```
<!-- small cluster memory settings -->
<property>
  <name>mapreduce.map.memory.mb</name>
  <value>256</value>
</property>

<property>
  <name>mapreduce.reduce.memory.mb</name>
  <value>256</value>
</property>

<property>
  <description>Application master allocation</description>
  <name>yarn.app.mapreduce.am.resource.mb</name>
  <value>256</value>
</property>

<property>
  <name>mapreduce.map.java.opts</name>
  <value>-Xmx204m</value>
</property>

<property>
  <name>mapreduce.reduce.java.opts</name>
  <value>-Xmx204m</value>
</property>

<property>
  <description>Application Master JVM opts</description>
  <name>yarn.app.mapreduce.am.command-opts</name>
  <value>-Xmx204m</value>
</property>

<property>
  <name>mapreduce.task.io.sort.mb</name>
  <value>50</value>
</property>
```

# Result

```
2024-12-01 15:34:16,639 INFO mapreduce.Job: map 0% reduce 0%
2024-12-01 15:34:38,261 INFO mapreduce.Job: map 100% reduce 0%
2024-12-01 15:34:47,552 INFO mapreduce.Job: map 100% reduce 100%
2024-12-01 15:34:48,640 INFO mapreduce.Job: Job job_1733067204611_0001
completed successfully
```

```
2024-12-01 15:34:48,828 INFO mapreduce.Job: Counters: 54
```

## File System Counters

```
FILE: Number of bytes read=79
FILE: Number of bytes written=928312
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=280
HDFS: Number of bytes written=41
HDFS: Number of read operations=11
HDFS: Number of large read operations=0
HDFS: Number of write operations=2
HDFS: Number of bytes read erasure-coded=0
```

## Job Counters

```
Launched map tasks=2
Launched reduce tasks=1
Data-local map tasks=2
Total time spent by all maps in occupied slots (ms)=36550
Total time spent by all reduces in occupied slots (ms)=6912
Total time spent by all map tasks (ms)=36550
Total time spent by all reduce tasks (ms)=6912
Total vcore-milliseconds taken by all map tasks=36550
Total vcore-milliseconds taken by all reduce tasks=6912
Total megabyte-milliseconds taken by all map tasks=9356800
Total megabyte-milliseconds taken by all reduce tasks=1769472
```

## Map-Reduce Framework

```
Map input records=2
Map output records=8
Map output bytes=82
Map output materialized bytes=85
Input split bytes=230
Combine input records=8
Combine output records=6
Reduce input groups=5
Reduce shuffle bytes=85
Reduce input records=6
Reduce output records=5
Spilled Records=12
Shuffled Maps =2
Failed Shuffles=0
Merged Map outputs=2
GC time elapsed (ms)=526
CPU time spent (ms)=3030
Physical memory (bytes) snapshot=561733632
Virtual memory (bytes) snapshot=5992550400
Total committed heap usage (bytes)=293806080
Peak Map Physical memory (bytes)=213331968
Peak Map Virtual memory (bytes)=1995571200
Peak Reduce Physical memory (bytes)=146944000
Peak Reduce Virtual memory (bytes)=2003013632
```

## Shuffle Errors

```
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
```

## File Input Format Counters

```
Bytes Read=50
```

## File Output Format Counters

```
Bytes Written=41
```

```
HDP.tlp - hdadmin@192.168.18.122:22 - Bitvise xterm - hdadmin@master: ~/sour...
Peak Map Virtual memory (bytes)=1995571200
Peak Reduce Physical memory (bytes)=146944000
Peak Reduce Virtual memory (bytes)=2003013632

Shuffle Errors
BAD_ID=0
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
Bytes Read=50

File Output Format Counters
Bytes Written=41

hdadmin@master:~/source_code$ hadoop fs -ls /user/joe/wordcount/output2
2024-12-01 15:36:21,143 WARN util.NativeCodeLoader: Unable to load native-
hadoop library for your platform... using builtin-java classes where appli-
cable
Found 2 items
-rw-r--r-- 1 hdadmin supergroup 0 2024-12-01 15:34 /user/joe/wo
rdcount/output2/_SUCCESS
-rw-r--r-- 1 hdadmin supergroup 41 2024-12-01 15:34 /user/joe/wo
rdcount/output2/part-r-00000
hdadmin@master:~/source_code$ hadoop fs -cat /user/joe/wordcount/output2/p
art-r-00000
2024-12-01 15:37:08,464 WARN util.NativeCodeLoader: Unable to load native-
hadoop library for your platform... using builtin-java classes where appli-
cable
Bye 1
Goodbye 1
Hadoop 2
Hello 2
World 2
hdadmin@master:~/source_code$
```

# Word Count Job Status (Success)



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Tools

Cluster Metrics

| Apps Submitted | Apps Pending | Apps Running | Apps Completed | Containers Running |
|----------------|--------------|--------------|----------------|--------------------|
| 1              | 0            | 0            | 1              | 0                  |

Cluster Nodes Metrics

| Active Nodes | Decommissioning Nodes | Decommissioned |
|--------------|-----------------------|----------------|
| 2            | 0                     | 0              |

Scheduler Metrics

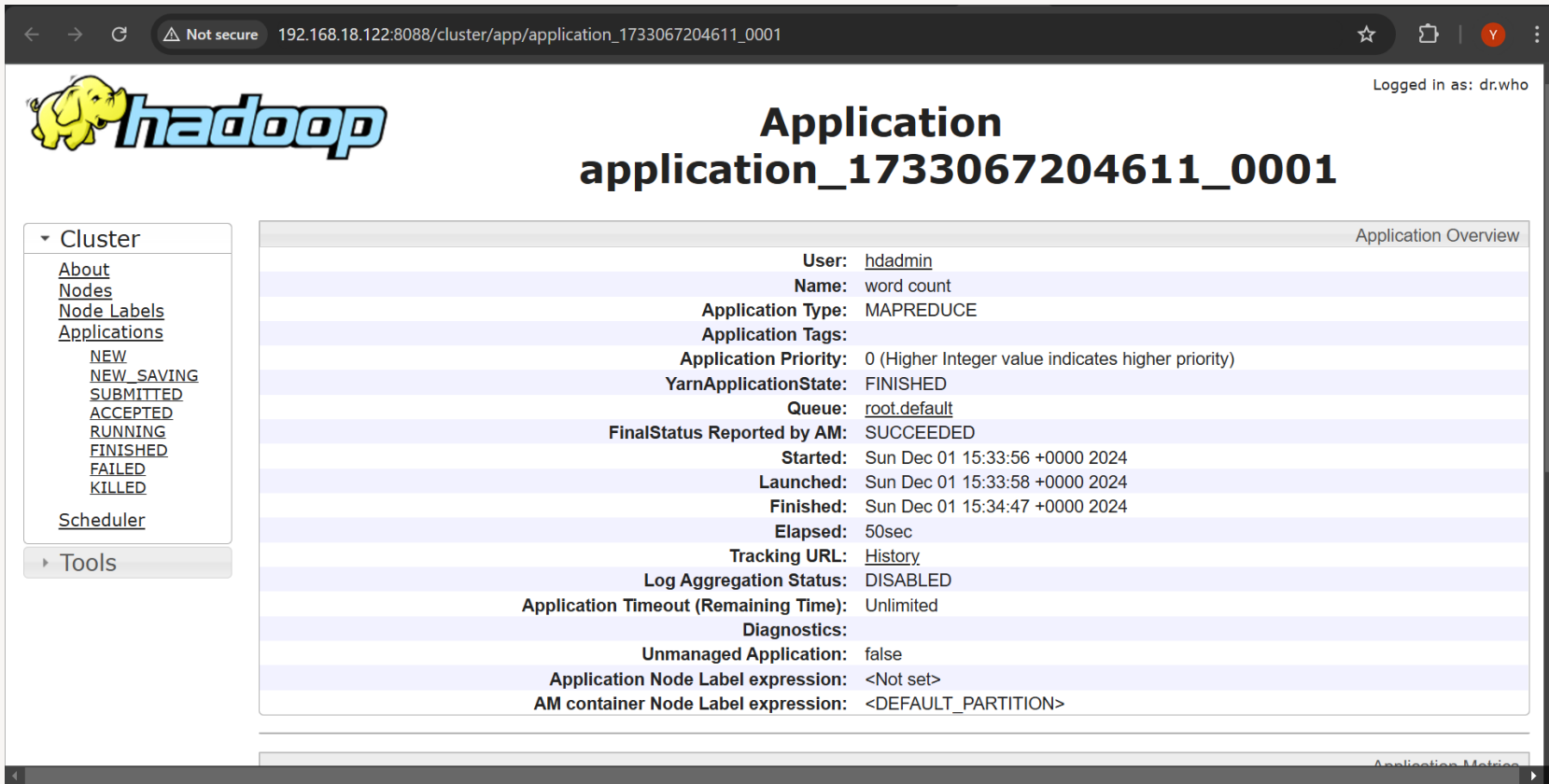
| Scheduler Type     | Scheduling Resource Type     | Minimum Allocation     | Maximum Allocation      |
|--------------------|------------------------------|------------------------|-------------------------|
| Capacity Scheduler | [memory-mb (unit=M), vcores] | <memory:256, vCores:1> | <memory:4096, vCores:1> |

Show 20 entries

| ID                                             | User    | Name       | Application Type | Application Tags | Queue        | Application Priority | StartTime                     | LaunchTime                    |
|------------------------------------------------|---------|------------|------------------|------------------|--------------|----------------------|-------------------------------|-------------------------------|
| <a href="#">application_1733067204611_0001</a> | hdadmin | word count | MAPREDUCE        |                  | root.default | 0                    | Sun Dec 1 22:33:56 +0700 2024 | Sun Dec 1 22:33:58 +0700 2024 |

Showing 1 to 1 of 1 entries

# Application Status



The screenshot shows the Hadoop YARN web interface. The browser address bar indicates the URL is 192.168.18.122:8088/cluster/app/application\_1733067204611\_0001. The page title is "Application application\_1733067204611\_0001". The user is logged in as dr.who. The left sidebar contains a "Cluster" menu with links to About, Nodes, Node Labels, Applications, and Scheduler. The "Applications" link is selected, showing a list of application states: NEW, NEW\_SAVING, SUBMITTED, ACCEPTED, RUNNING, FINISHED, FAILED, and KILLED. The "Tools" menu is also visible. The main content area displays the "Application Overview" for the selected application. The overview includes the following details:

| Application Overview                  |                                                    |
|---------------------------------------|----------------------------------------------------|
| User:                                 | hdadmin                                            |
| Name:                                 | word count                                         |
| Application Type:                     | MAPREDUCE                                          |
| Application Tags:                     |                                                    |
| Application Priority:                 | 0 (Higher Integer value indicates higher priority) |
| YarnApplicationState:                 | FINISHED                                           |
| Queue:                                | root.default                                       |
| FinalStatus Reported by AM:           | SUCCEEDED                                          |
| Started:                              | Sun Dec 01 15:33:56 +0000 2024                     |
| Launched:                             | Sun Dec 01 15:33:58 +0000 2024                     |
| Finished:                             | Sun Dec 01 15:34:47 +0000 2024                     |
| Elapsed:                              | 50sec                                              |
| Tracking URL:                         | History                                            |
| Log Aggregation Status:               | DISABLED                                           |
| Application Timeout (Remaining Time): | Unlimited                                          |
| Diagnostics:                          |                                                    |
| Unmanaged Application:                | false                                              |
| Application Node Label expression:    | <Not set>                                          |
| AM container Node Label expression:   | <DEFAULT_PARTITION>                                |

# Hadoop MapReduce (Kill Stuck MR Jobs)

```
./yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.1.jar pi 16 1
```

```
mapred job -list  
mapred job -kill <Job_ID>
```

```
yarn application -list  
yarn application -kill <Application_ID>
```

# YARN Administrative Tools

yarn radmin -help

[Adding and Decommission Nodes]

yarn radmin -refreshNodes

[Capacity Scheduler / capacity-scheduler.xml]

yarn radmin -refreshQueues

[User to Group Mapping]

yarn radmin -refreshUserToGroupMappings

[Resource Manager Admin ACL]

yarn radmin -refreshAdminAcls

yarn radmin -refreshServiceAcl

```
<property>  
  <name>yarn.resourcemanager.admin.acl</name>  
  <value>hdadmin, hadoop</value>  
  <description>List of users or groups who are allowed to perform administrative actions on the ResourceManager</description>  
</property>
```

# Resource Manager Job History (1 Nodes)

```
<!-- The address where the JobHistory server listens for job history client requests. -->
<property>
  <name>mapreduce.jobhistory.address</name>
  <value>master:10020</value>
  <description>The host and port where the JobHistory server listens for requests.</description>
</property>

<!-- The address for the web interface of the JobHistory server to view job history. -->
<property>
  <name>mapreduce.jobhistory.webapp.address</name>
  <value>master:19888</value>
  <description>The host and port where the JobHistory server web interface is available.</description>
</property>

<!-- The secure address for the web interface of the JobHistory server to view job history. . -->
<property>
  <name>mapreduce.jobhistory.webapp.https.address</name>
  <value>master:19890</value>
</property>

<!-- The local directory where job history logs are stored on the Job History Server node. -->
<property>
  <name>mapreduce.jobhistory.done-dir</name>
  <value>/usr/local/hadoop_store/mapreduce/done</value>
  <description>Local directory for storing job history log files.</description>
</property>

<property>
  <name>mapreduce.jobhistory.intermediate-done-dir</name>
  <value>/usr/local/hadoop_store/mapreduce/done_intermediate</value>
  <description>Local directory for storing job history log files.</description>
</property>

<!-- Maximum retention time for job history in milliseconds (e.g., 30 days). -->
<property>
  <name>mapreduce.jobhistory.cleaner.interval-ms</name>
  <value>2592000000</value> <!-- 30 days in milliseconds -->
</property>
```

# Localizer Library (NodeManager)

```
<!-- Localizer listens on port 8040, can be changed if needed -->
<property>
  <name>yarn.nodemanager.localizer.address</name>
  <value>0.0.0.0:8040</value>
  <description>The address where the NodeManager localizer listens for requests</description>
</property>

<!-- The localizer cache size -->
<property>
  <name>yarn.nodemanager.localizer.cache.target-size-mb</name>
  <value>2048</value>
  <description>The target size of the localizer cache</description>
</property>

<!-- Cache cleanup interval -->
<property>
  <name>yarn.nodemanager.localizer.cache.cleanup.interval-ms</name>
  <value>600000</value>
  <description>The interval (in milliseconds) to check for updates in the localizer cache</des>
</property>
```

[Node Manager]

./bin/yarn --daemon start nodemanager

# Change Resource Manager Scheduler

[yarn-site.xml]

```
<property>
  <name>yarn.resourcemanager.scheduler.class</name>
<value>org.apache.hadoop.yarn.server.resourcemanager.scheduler.capacity.CapacityScheduler</value>
</property>

<property>
  <name>yarn.resourcemanager.scheduler.class</name>
<value>org.apache.hadoop.yarn.server.resourcemanager.scheduler.fair.FairScheduler</value>
</property>
```

[Start Job With Specific Queue in Yarn]

```
yarn jar my-job.jar -Dmapreduce.job.queueName=<queue_name>
```

# capacity-scheduler.xml (Queue)

```
<configuration>
  <!-- Define root-level queues -->
  <property>
    <name>yarn.scheduler.capacity.root.queues</name>
    <value>default,app1,app2</value> <!-- Top-level queues: default, app1, app2 -->
  </property>

  <!-- Define capacity for default queue -->
  <property>
    <name>yarn.scheduler.capacity.root.default.capacity</name>
    <value>50</value> <!-- 50% capacity for default -->
  </property>

  <!-- Define capacity for app1 queue -->
  <property>
    <name>yarn.scheduler.capacity.root.app1.capacity</name>
    <value>25</value> <!-- 25% capacity for app1 -->
  </property>

  <!-- Define capacity for app2 queue -->
  <property>
    <name>yarn.scheduler.capacity.root.app2.capacity</name>
    <value>25</value> <!-- 25% capacity for app2 -->
  </property>

</configuration>
```

# fair-scheduler.xml

```
<?xml version="1.0"?>
<allocations>
  <user name="hdadmin">
    <maxRunningJobs>100</maxRunningJobs>
  </user>
  <queuePlacementPolicy>
    <rule name="default" queue="low"/>
  </queuePlacementPolicy>
  <userMaxAppsDefault>10</userMaxAppsDefault>
  <queue name="high">
    <minResources>4096 mb, 1 vcores</minResources>
    <schedulingMode>fair</schedulingMode>
    <weight>0.5</weight>
    <minSharePreemptionTimeout>2</minSharePreemptionTimeout>
  </queue>
  <queue name="normal">
    <minResources>3072 mb, 1 vcores</minResources>
    <schedulingMode>fair</schedulingMode>
    <weight>0.3</weight>
    <minSharePreemptionTimeout>2</minSharePreemptionTimeout>
  </queue>
  <queue name="low">
    <minResources>1024 mb, 1 vcores</minResources>
    <weight>0.2</weight>
    <schedulingMode>fair</schedulingMode>
    <minSharePreemptionTimeout>2</minSharePreemptionTimeout>
  </queue>
</allocations>
```

# yarn-site.xml (Fair Scheduler)

```
<configuration>
  <!-- Path to the fair scheduler allocation file -->
  <property>
    <name>yarn.scheduler.fair.allocation.file</name>
    <value>/usr/local/hadoop/etc/hadoop/fair-scheduler.xml</value>
    <description>Path to the fair scheduler allocation file that defines queues and their resource allocation.</description>
  </property>
  <property>
    <name>yarn.scheduler.fair.user-as-default-queue</name>
    <value>true</value>
  </property>
  <!-- Enable or disable preemption -->
  <property>
    <name>yarn.scheduler.fair.preemption</name>
    <value>true</value>
    <description>If true, preemption is enabled to reclaim resources for jobs that are starving.</description>
  </property>
  <property>
    <name>yarn.scheduler.fair.preemption.cluster-utilization-threshold</name>
    <value>0.8f</value>
  </property>
  <property>
    <name>yarn.scheduler.fair.sizebasedweight</name>
    <value>true</value>
  </property>
  <property>
    <name>yarn.scheduler.fair.assignmultiple</name>
    <value>true</value>
  </property>
</configuration>
```

# yarn-site.xml (Fair Scheduler)

```
<property>
  <name>yarn.scheduler.fair.dynamic.max.assign</name>
  <value>>false</value>
</property>
<property>
  <name>yarn.scheduler.fair.max.assign</name>
  <value>-1</value>
</property>
<property>
  <name>yarn.scheduler.fair.locality.threshold.node</name>
  <value>-1.0</value>
</property>
<property>
  <name>yarn.scheduler.fair.locality.threshold.rack</name>
  <value>-1.0</value>
</property>
<property>
  <name>yarn.scheduler.fair.allow-undeclared-pools</name>
  <value>true</value>
</property>
<property>
  <name>yarn.scheduler.fair.update-interval-ms</name>
  <value>500</value>
</property>
<property>
  <name>yarn.resource-types.memory-mb.increment-allocation</name>
  <value>256</value>
</property>
<property>
  <name>yarn.resource-types.vcores.increment-allocation</name>
  <value>1</value>
</property>
</configuration>
```

# Result (Fair Scheduler)

Cluster

About

Nodes

Node Labels

Applications

NEW

NEW SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Nodes of the cluster

Cluster Metrics

| Apps Submitted | Apps Pending | Apps Running | Apps Completed | Containers Running | Used Resources         | Total Resources         | Reserved Resources     | Physical Mem Used | Physical VCores Used |
|----------------|--------------|--------------|----------------|--------------------|------------------------|-------------------------|------------------------|-------------------|----------------------|
| 1              | 0            | 0            | 1              | 0                  | <memory:0 B, vCores:0> | <memory:8 GB, vCores:2> | <memory:0 B, vCores:0> | 21                | 0                    |

Cluster Nodes Metrics

| Active Nodes | Decommissioning Nodes | Decommissioned Nodes | Lost Nodes | Unhealthy Nodes | Rebooted Nodes | Shutdown Nodes |
|--------------|-----------------------|----------------------|------------|-----------------|----------------|----------------|
| 2            | 0                     | 0                    | 0          | 0               | 0              | 0              |

User Metrics for dr.who

| Apps Submitted | Apps Pending | Apps Running | Apps Completed | Containers Running | Containers Pending | Containers Reserved | Memory Used | Memory Pending | Memory Reserved | VCores Used | VCores Pending | VCores Reserved |
|----------------|--------------|--------------|----------------|--------------------|--------------------|---------------------|-------------|----------------|-----------------|-------------|----------------|-----------------|
| 0              | 0            | 0            | 0              | 0                  | 0                  | 0                   | 0 B         | 0 B            | 0 B             | 0           | 0              | 0               |

Scheduler Metrics

| Scheduler Type | Scheduling Resource Type      | Minimum Allocation     | Maximum Allocation      | Maximum Cluster Application Priority | Scheduler Busy % | RM Dispatcher EventQueue Size | Scheduler Dispatcher EventQueue Size |
|----------------|-------------------------------|------------------------|-------------------------|--------------------------------------|------------------|-------------------------------|--------------------------------------|
| Fair Scheduler | [memory-mb (unit-Mi), vcores] | <memory:256, vCores:1> | <memory:4096, vCores:1> | 0                                    | 0                | 0                             | 0                                    |

Show 20 entries

| Node Labels | Rack          | Node State | Node Address | Node HTTP Address | Last health-update             | Health-report | Containers | Allocation Tags | Mem Used | Mem Avail | Phys Mem Used % | VCores Used | VCores Avail | Phys VCores Used % | Version |
|-------------|---------------|------------|--------------|-------------------|--------------------------------|---------------|------------|-----------------|----------|-----------|-----------------|-------------|--------------|--------------------|---------|
|             | /default-rack | RUNNING    | slave2:41037 | slave2:8042       | Wed Dec 04 10:13:00 +0000 2024 |               | 0          |                 | 0 B      | 4 GB      | 22              | 0           | 1            | 0                  | 3.4.1   |
|             | /default-rack | RUNNING    | slave1:45715 | slave1:8042       | Wed Dec 04 10:13:01 +0000 2024 |               | 0          |                 | 0 B      | 4 GB      | 20              | 0           | 1            | 0                  | 3.4.1   |

```
2024-12-04 10:09:47,075 INFO impl.YarnClientImpl: Submitted application application_1733306959569_0001
2024-12-04 10:09:47,125 INFO mapreduce.Job: The url to track the job: http://master:8088/proxy/application_1733306959569_0001/
2024-12-04 10:09:47,127 INFO mapreduce.Job: Running job: job_1733306959569_0001
2024-12-04 10:10:07,948 INFO mapreduce.Job: Job job_1733306959569_0001 running in uber mode : false
2024-12-04 10:10:07,951 INFO mapreduce.Job: map 0% reduce 0%
2024-12-04 10:10:20,384 INFO mapreduce.Job: map 50% reduce 0%
2024-12-04 10:10:31,604 INFO mapreduce.Job: map 100% reduce 0%
2024-12-04 10:10:43,841 INFO mapreduce.Job: map 100% reduce 100%
2024-12-04 10:10:44,923 INFO mapreduce.Job: Job job_1733306959569_0001 completed successfully
2024-12-04 10:10:45,269 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=79
  FILE: Number of bytes written=939619
  FILE: Number of read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=280
  HDFS: Number of bytes written=41
  HDFS: Number of read operations=11
  HDFS: Number of large read operations=0
  HDFS: Number of write operations=2
  HDFS: Number of bytes read erasure-coded=0

Job Counters
  Launched map tasks=2
  Launched reduce tasks=1
  Data-local map tasks=2
  Total time spent by all maps in occupied slots (ms)=19746
  Total time spent by all reduces in occupied slots (ms)=10219
  Total time spent by all map tasks (ms)=19746
  Total time spent by all reduce tasks (ms)=10219
  Total vcore-millisecods taken by all map tasks=19746
  Total vcore-millisecods taken by all reduce tasks=10219
  Total megabyte-millisecods taken by all map tasks=5954976
  Total megabyte-millisecods taken by all reduce tasks=2616064

Map-Reduce Framework
```

hadoop jar wc.jar WordCount /user/joe/wordcount/input /user/joe/wordcount/output3

hadoop jar wc.jar WordCount /user/joe/wordcount/input /user/joe/wordcount/output3 -Dmapreduce.job.queueName=low -Dmapreduce.job.priority=LOW

# MapReduce Example Jobs

```
$HADOOP_HOME/bin/yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.1.jar teragen 1000000000 <HDFS_PATH>
```

```
$HADOOP_HOME/bin/yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.1.jar terasort <HDFS_INPUT> <HDFS_OUTPUT>
```

```
$HADOOP_HOME/bin/yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-examples-3.4.1.jar teravalidate /user/joe/wordcount/output7 /output8
```

```
$HADOOP_HOME/bin/yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-jobclient-3.4.1-tests.jar TestDFSIO -write -nrFiles 10 -fileSize 1000
```

```
$HADOOP_HOME/bin/yarn jar $HADOOP_HOME/share/hadoop/mapreduce/hadoop-mapreduce-client-jobclient-3.4.1-tests.jar TestDFSIO -clean
```

# Install Spark Scala

```
sudo apt-get install scala
wget https://dlcdn.apache.org/spark/spark-3.5.3/spark-3.5.3-bin-hadoop3.tgz
tar -xvzf spark-3.5.3-bin-hadoop3.tgz
mv spark-3.5.3-bin-hadoop3 spark
sudo chown -R hdadmin:hadoop ~/spark
sudo chown -R hdadmin:hadoop /usr/local/spark
```

```
[.bashrc]
export SPARK_HOME=/usr/local/spark
export SPARK_LOG_DIR=/app/spark/log
export SPARK_WORKER_DIR=/app/spark/tmp
export PATH=$PATH:$SPARK_HOME/bin
export SPARK_MASTER_HOST=$(grep master-node /etc/hosts | tail -n 1 | awk '{print $1}')
```

```
[Master Node]
export SPARK_MASTER_HOST=$(grep master-node /etc/hosts | tail -n 1 | awk '{print $1}')
sudo scp -r /usr/local/spark/ hdadmin@slave1:~/spark /
sudo scp -r /usr/local/spark/ hdadmin@slave2:~/spark/
```

```
hdadmin@master:/usr/local/spark$ jps
1666 ResourceManager
1204 NameNode
4856 Master
2554 ApplicationHistoryServer
4954 Jps
1452 SecondaryNameNode
```

```
hdadmin@slave1:~$ jps
4272 Jps
1089 DataNode
1239 NodeManager
4158 Worker
```

```
hdadmin@slave2:~$ jps
4368 Worker
1173 NodeManager
4493 Jps
1023 DataNode
```

# Result Spark Scala

←

→

↺

⚠ Not secure

192.168.18.125:8080

☆

📁

Y

⋮

3.5.3

Spark Master at spark://master:7077

**URL:** spark://master:7077

**Alive Workers:** 2

**Cores in use:** 2 Total, 0 Used

**Memory in use:** 5.6 GiB Total, 0.0 B Used

**Resources in use:**

**Applications:** 0 Running, 0 Completed

**Drivers:** 0 Running, 0 Completed

**Status:** ALIVE

▼ Workers (2)

| Worker Id                                                  | Address              | State | Cores      | Memory               | Resources |
|------------------------------------------------------------|----------------------|-------|------------|----------------------|-----------|
| <a href="#">worker-20241206175128-192.168.18.123-40955</a> | 192.168.18.123:40955 | ALIVE | 1 (0 Used) | 2.8 GiB (0.0 B Used) |           |
| <a href="#">worker-20241206175128-192.168.18.124-33383</a> | 192.168.18.124:33383 | ALIVE | 1 (0 Used) | 2.8 GiB (0.0 B Used) |           |

▼ Running Applications (0)

| Application ID | Name | Cores | Memory per Executor | Resources Per Executor | Submitted Time | User | State | Duration |
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|

▼ Completed Applications (0)

| Application ID | Name | Cores | Memory per Executor | Resources Per Executor | Submitted Time | User | State | Duration |
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|
|----------------|------|-------|---------------------|------------------------|----------------|------|-------|----------|

# Test Job (Normal & YARN Example)

[Normal]

`$SPARK_HOME/bin/run-example SparkPi 5`

`$SPARK_HOME/bin/spark-submit examples/src/main/python/pi.py 5`

[YARN]

`$SPARK_HOME/bin/spark-submit --class org.apache.spark.examples.SparkPi --master yarn --deploy-mode cluster --driver-memory 1g --executor-memory 1g --executor-cores 1 --queue low examples/jars/spark-examples*.jar`



Cluster

About

Nodes

Node Labels

Applications

NEW

NEW

SAVING

SUBMITTED

ACCEPTED

RUNNING

FINISHED

FAILED

KILLED

Scheduler

Tools

Cluster Metrics

6 Apps Submitted0 Apps Pending1 Apps Running5 Apps Completed1 Containers Running<memory: 1.50 GB, vCores: 1>Used Resources<memory: 8 GB, vCores: 2>Total Resources

Cluster Nodes Metrics

2 Active Nodes0 Decommissioning Nodes0 Decommissioned Nodes0 Lost Nodes0

User Metrics for dr:who

0 Apps Submitted0 Apps Pending0 Apps Running0 Apps Completed0 Containers Running0 Containers Pending0 Containers Reserved0 B Memory Used

Scheduler Metrics

Scheduler TypeScheduling Resource TypeMinimum AllocationMaximum AllocationMaximum Cluster Application Priority

Fair Scheduler[memory-mb (unit=M), vcores]<memory: 256, vCores: 1><memory: 4096, vCores: 1>00

Show 20 entries

ID

User

Name

Application Type

Application Tags

Queue

Application Priority

StartTime

LaunchTime

FinishTime

State

FinalStatus

Running Containers

Allocate CPU Vcores

application\_1733487429361\_0006

hdadmin

org.apache.spark.examples.SparkPi

SPARK

root.low

0

Sat Dec 7 01:21:54 +0700 2024

Sat Dec 7 01:21:55 +0700 2024

N/A

RUNNING

UNDEFINED

1

1

Showing 1 to 1 of 1 entries

```
HDP:slp - hdadmin@192.168.18.125:22 - Bitvise xterm - hdadmin@master: /usr/local/spark
24/12/06 18:09:27 INFO Client:
client token: N/A
diagnostics: N/A
ApplicationMaster host: slave2
ApplicationMaster RPC port: 36031
queue: root.low
start time: 1733508546281
final status: UNDEFINED
tracking URL: http://master:8088/proxy/application_1733487429361_0005/
user: hdadmin
24/12/06 18:09:57 INFO Client: Application report for application_1733487429361_0005 (state: RUNNING)
24/12/06 18:09:58 INFO Client: Application report for application_1733487429361_0005 (state: FINISHED)
24/12/06 18:09:58 INFO Client:
client token: N/A
diagnostics: N/A
ApplicationMaster host: slave2
ApplicationMaster RPC port: 36031
queue: root.low
start time: 1733508546281
final status: SUCCEEDED
tracking URL: http://master:8088/proxy/application_1733487429361_0005/
user: hdadmin
24/12/06 18:09:58 INFO Client: Deleted staging directory hdfs://master:8020/user/hdadmin/.sparkStaging/application_1733487429361_0005
24/12/06 18:09:58 INFO ShutdownHookManager: Shutdown hook called
24/12/06 18:09:58 INFO ShutdownHookManager: Deleting directory /tmp/spark-5704fa8d-dfc2-4656-afea-e95a254ef45b
24/12/06 18:09:58 INFO ShutdownHookManager: Deleting directory /tmp/spark-94d5cf42-8b75-4bd2-ad86-1ebb0d13f2b5
hdadmin@master: /usr/local/spark$
```

# Hadoop FSCK & DFSADMIN Command

```
HDP:tlp - hdadmin@192.168.18.125:22 - Bitwise xterm - hdadmin@master: /usr/local/hadoop
hdadmin@master:/usr/local/hadoop$ mkdir conf
hdadmin@master:/usr/local/hadoop$ nano topology.py
hdadmin@master:/usr/local/hadoop$ hdfs fsck /
2024-12-06 18:31:05,907 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Connecting to namenode via http://master:9870/fsck?ugi=hdadmin&path=%2F
FSCK started by hdadmin (auth:SIMPLE) from /192.168.18.125 for path / at Fri Dec 06 18:31:07 UTC 2024

Status: HEALTHY
Number of data-nodes: 2
Number of racks: 1
Total dirs: 4
Total symlinks: 0

Replicated Blocks:
Total size: 0 B
Total files: 0
Total blocks (validated): 0
Minimally replicated blocks: 0
Over-replicated blocks: 0
Under-replicated blocks: 0
Mis-replicated blocks: 0
Default replication factor: 1
Average block replication: 0.0
Missing blocks: 0
Corrupt blocks: 0
Missing replicas: 0
Blocks queued for replication: 0

Erasure Coded Block Groups:
Total size: 0 B
Total files: 0
```

`$HADOOP_HOME/bin/hdfs fsck <HDFS_PATH>`

`$HADOOP_HOME/bin/hdfs dfsadmin -report`

`$HADOOP_HOME/bin/hdfs dfsadmin -printTopology`

[Empty Trash Folder]

`hadoop fs -expunge`

[Other FSCK Command]

`hdfs fsck -list-corruptfileblocks`

`hdfs fsck -delete`

`hdfs fsck / -files -blocks -locations -racks`

`hdfs fsck -move`

[Other DFSADMIN Command]

`hdfs dfsadmin -refreshNodes`

`hdfs dfsadmin -metasave <HDFS_FILENAME>`

# HDFS DFS & Balancer

```
hdfs dfs -df -h
```

```
hdfs dfs -du <HDFS_PATH>
```

[In Percentage]

```
hdfs balancer
```

```
hdfs balancer -threshold 3
```

# Install MySQL JDBC Connector (All Nodes)

```
wget https://dev.mysql.com/get/Downloads/Connector-J/mysql-connector-j_9.1.0-1ubuntu22.04_all.deb  
sudo dpkg -i mysql-connector-j_9.1.0-1ubuntu22.04_all.deb
```

```
hdadmin@master:~$ sudo dpkg -i mysql-connector-j_9.1.0-1ubuntu22.04_all.deb  
Selecting previously unselected package mysql-connector-j.  
(Reading database ... 87084 files and directories currently installed.)  
Preparing to unpack mysql-connector-j_9.1.0-1ubuntu22.04_all.deb ...  
Unpacking mysql-connector-j (9.1.0-1ubuntu22.04) ...  
Setting up mysql-connector-j (9.1.0-1ubuntu22.04) ...  
hdadmin@master:~$
```

# Secondary, Checkpoint, Backup, Standby

Secondary NameNode → Perform periodic checkpoint of FSImage and doesn't upload the checkpointed file to active NameNode

Checkpoint Node → Same as Secondary NameNode, but upload the new FSImage file to active NameNode

Backup Node → Save the namespace to current local and FSImage file and zeroes the edit log file, giving the newest version. No need of Secondary or Checkpoint. Remove config dfs.namenode.name.dir on hdfs-site.xml

hdfs namenode -backup

hdfs namenode -importCheckpoint

Standby NameNode → The active-passive hot standby node for NameNode. Main function as failover and other NameNodes job (checkpoint, upload, etc)

# Configure High Availability NameNode (QJM)

<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HDFSHighAvailabilityWithNFS.html>  
<https://hadoop.apache.org/docs/stable/hadoop-project-dist/hadoop-hdfs/HDFSHighAvailabilityWithQJM.html>

[hdfs-site.xml]

```
<!-- Logical Name -->
<property>
  <name>dfs.nameservices</name>
  <value>mycluster</value>
</property>
<!-- Which node is NameNodes -->
<property>
  <name>dfs.ha.namenodes.mycluster</name>
  <value>nn1,nn2</value>
</property>

<!-- RPC -->
<property>
  <name>dfs.namenode.rpc-address.mycluster.nn1</name>
  <value>slave1:8020</value>
</property>
<property>
  <name>dfs.namenode.rpc-address.mycluster.nn2</name>
  <value>slave2:8020</value>
</property>
```

[hdfs-site.xml]

```
<!-- HTTP Address -->
<property>
  <name>dfs.namenode.http-address.mycluster.nn1</name>
  <value>slave1:9870</value>
</property>
<property>
  <name>dfs.namenode.http-address.mycluster.nn2</name>
  <value>slave2:9870</value>
</property>

<!-- Shared FSImage -->
<property>
  <name>dfs.namenode.shared.edits.dir</name>
  <value>qjournal://slave1:8485;slave2:8485;master:8485/mycluster</value>
</property>
<property>
  <name>dfs.namenode.edits.qjournals.resolution-enabled</name>
  <value>true</value>
</property>
<property>
  <name>dfs.client.failover.proxy.provider.mycluster</name>

<value>org.apache.hadoop.hdfs.server.namenode.ha.ConfiguredFailoverProxyProvider</v
alue>
</property>
```

# Configure High Availability NameNode (QJM)

[hdfs-site.xml]

```
<!-- SSH Fencing -->
<property>
  <name>dfs.ha.fencing.methods</name>
  <value>sshfence</value>
</property>

<property>
  <name>dfs.ha.fencing.ssh.private-key-files</name>
  <value>/home/hdadmin/.ssh/id_rsa</value>
</property>

<property>
  <name>dfs.ha.fencing.ssh.connect-timeout</name>
  <value>30000</value>
</property>

<!-- Local State Journal Node Edit Log Path -->
<property>
  <name>dfs.journalnode.edits.dir</name>
  <value>/usr/local/hadoop_store/hdfs/journalnode</value>
</property>

<!-- Disable Write On Safe Mode (Prevent Conflict Write) -->
<property>
  <name>dfs.ha.nn.not-become-active-in-safemode</name>
  <value>true</value>
</property>
```

Presentation title

[core-site.xml]

```
<property>
  <name>fs.defaultFS</name>
  <value>hdfs://mycluster</value>
</property>
```

[hdfs-site.xml]

```
<!-- Zookeeper Quorum -->
<property>
  <name>ha.zookeeper.quorum</name>
  <value>master:2181,slave1:2181,slave2:2181</value>
</property>
```

[All Nodes]

```
sudo mkdir /usr/local/hadoop_store/hdfs/journalnode
sudo chown -R hdadmin:hadoop /usr/local/hadoop_store/hdfs/journalnode
```

[hdfs-site.xml]

```
<!-- Automatic Failover -->
<property>
  <name>dfs.ha.automatic-failover.enabled</name>
  <value>true</value>
</property>
```

# Quorum Journal Manager (Results)

```
hdadmin@master:/usr/local/hadoop/etc/hadoop$ ../../sbin/start-all.sh
WARNING: Attempting to start all Apache Hadoop daemons as hdadmin in 10 seconds.
WARNING: This is not a recommended production deployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [master slave1 slave2]
Starting datanodes
Starting journal nodes [slave2 slave1 master]
2024-12-07 09:21:16,007 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Starting ZK Failover Controllers on NN hosts [master slave1 slave2]
Starting resourcemanager
Starting nodemanagers
hdadmin@master:/usr/local/hadoop/etc/hadoop$ jps
14016 ResourceManager
13651 JournalNode
13384 NameNode
14330 Jps
hdadmin@master:/usr/local/hadoop/etc/hadoop$
```

```
Last login: Sat Dec 7 08:39:08 2024 from 192.168.18.217
hdadmin@slave1:~$ jps
5160 JournalNode
5034 DataNode
5322 NodeManager
5532 Jps
hdadmin@slave1:~$
```

```
Last login: Sat Dec 7 08:47:33 2024 from 192.168.18.217
hdadmin@slave2:~$ jps
5380 JournalNode
5253 DataNode
5542 NodeManager
5754 Jps
hdadmin@slave2:~$
```

# Install Zookeeper (Adding HA)

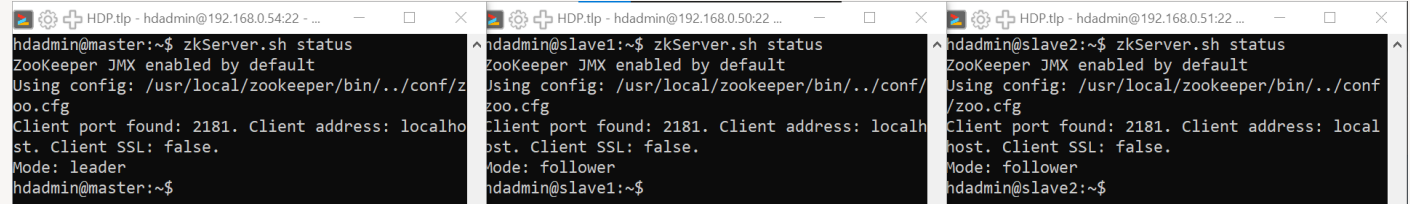
```
[.bashrc]
export ZOOKEEPER_HOME=/usr/local/zookeeper
export PATH=$PATH:$ZOOKEEPER_HOME/bin
```

```
[Latest Stable Release]
wget https://dlcdn.apache.org/zookeeper/zookeeper-3.8.4/apache-zookeeper-3.8.4-bin.tar.gz
tar -xvzf apache-zookeeper-3.8.4-bin.tar.gz
mv apache-zookeeper-3.8.4-bin zookeeper
```

```
cp $ZOOKEEPER_HOME/conf/zoo_sample.cfg $ZOOKEEPER_HOME/conf/zoo.cfg
```

```
[zoo.cfg]
dataDir=/usr/local/zookeeper/data
server.1=master:2888:3888
server.2=slave1:2889:3889
server.3=slave2:2890:3890
```

```
[All Nodes]
sudo chown -R hdadmin:hadoop ~/zookeeper
sudo chown -R hdadmin:hadoop /usr/local/zookeeper
sudo scp -r /usr/local/zookeeper/ hdadmin@slave1:~/zookeeper/
mkdir $ZOOKEEPER_HOME/data
nano myid (the value is a single unique number to nodes)
```



The image shows three terminal windows side-by-side, each displaying the output of the 'zkServer.sh status' command. The first window is on the master node (hdadmin@master), showing it is in 'leader' mode. The second window is on slave1 (hdadmin@slave1), showing it is in 'follower' mode. The third window is on slave2 (hdadmin@slave2), also showing it is in 'follower' mode. All three nodes report the same configuration: 'Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg', 'Client port found: 2181', and 'Client address: localhost'. The 'ZooKeeper JMX enabled by default' message is also present in all three.

```
hdadmin@master:~$ zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: leader
hdadmin@master:~$

hdadmin@slave1:~$ zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower
hdadmin@slave1:~$

hdadmin@slave2:~$ zkServer.sh status
ZooKeeper JMX enabled by default
Using config: /usr/local/zookeeper/bin/../conf/zoo.cfg
Client port found: 2181. Client address: localhost. Client SSL: false.
Mode: follower
hdadmin@slave2:~$
```

# High Availability Quorum Journal Manager

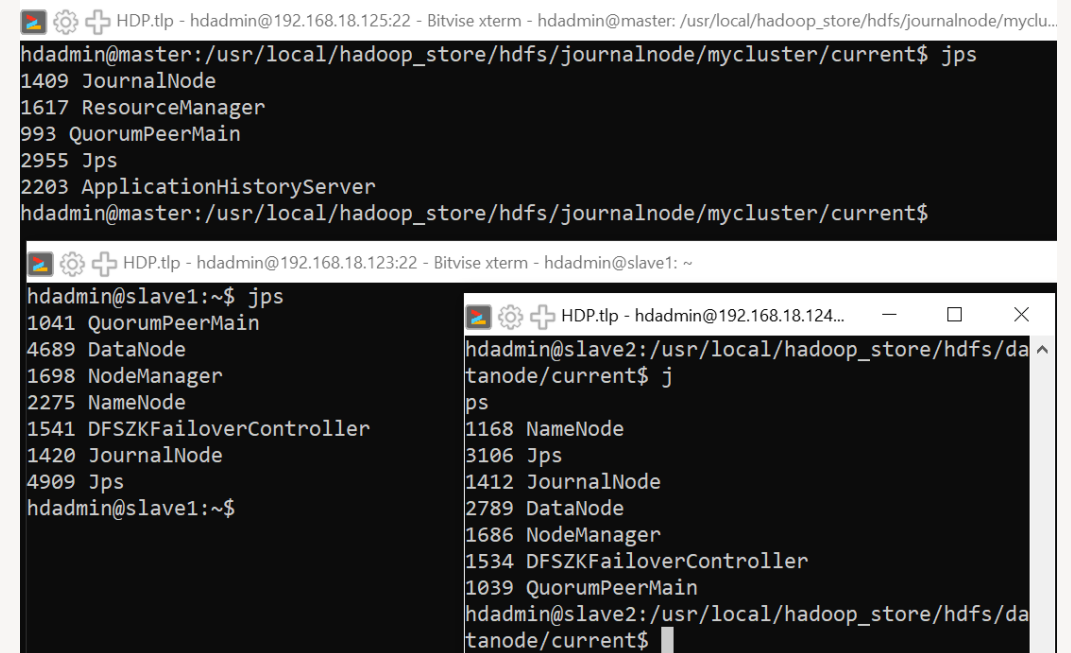
[All Nodes]  
zkServer.sh start

[Only Once]  
hdfs zkfc -formatZK -force  
hdfs --daemon start journalnode  
hdfs namenode -format  
hdfs namenode -format -clusterId mycluster -  
Ddfs.nameservices=mycluster  
hdfs namenode -initializeSharedEdits  
hdfs namenode -bootstrapStandby

[Clean Up]  
sudo rm -r /usr/local/hadoop\_store/\*

[All Nodes]  
sudo mkdir /app/hadoop/tmp/dfs/name  
sudo chown -R hdadmin:hadoop /app/hadoop/tmp/dfs/name

[Start All]  
\$HADOOP\_HOME/sbin/start-all.sh  
mapred --daemon start historyserver



The image shows three terminal windows. The top window is on the master node (hdadmin@master) running 'jps' and showing processes: JournalNode, ResourceManager, QuorumPeerMain, Jps, and ApplicationHistoryServer. The bottom-left window is on slave1 (hdadmin@slave1) running 'jps' and showing: QuorumPeerMain, DataNode, NodeManager, NameNode, DFSZKFailoverController, JournalNode, and Jps. The bottom-right window is on slave2 (hdadmin@slave2) running 'jps' and showing: NameNode, Jps, JournalNode, DataNode, NodeManager, DFSZKFailoverController, and QuorumPeerMain.

```
Starting namenodes on [slave1 slave2]
Starting datanodes
Starting journal nodes [slave2 slave1 master]
2024-12-07 19:18:52,938 WARN util.NativeCodeLoader: Unable to
atform... using builtin-java classes where applicable
Starting ZK Failover Controllers on NN hosts [slave1 slave2]
Starting resourcemanager
Starting nodemanagers
```

# Fix 2 HAQJM

```
<property>  
  <name>dfs.journalnode.rpc-address</name>  
  <value>slave1:8485</value>  
</property>
```

```
<property>  
  <name>dfs.journalnode.rpc-address</name>  
  <value>slave2:8485</value>  
</property>
```

```
<property>  
  <name>dfs.journalnode.rpc-address</name>  
  <value>master:8485</value>  
</property>
```

# Firewall Enable / Disable

[All Node]

```
sudo ufw enable
```

```
sudo ufw allow 22/tcp
```

```
sudo ufw allow 2181/tcp
```

```
sudo ufw allow 2888/tcp
```

```
sudo ufw allow 3888/tcp
```

# Administration Command (YARN)

```
yarn application -list --appStates <STATE>  
yarn application -status <Application_ID>  
yarn application -kill <Application_ID>  
yarn node -all -list  
yarn queue --status root.<QUEUE_NAME>
```

[Key]

```
yarn radmin -refreshNodes  
yarn radmin -transitionToActive  
yarn radmin -transitionToStandby  
yarn radmin -failover  
yarn radmin -getServiceState  
yarn radmin -checkHealth
```

# Decommissioning Node (All NameNodes)

```
echo "<hosts>" > $HADOOP_CONF_DIR/dfs.exclude  
echo "<hosts>" > $HADOOP_CONF_DIR/yarn.exclude
```

[hdfs-site.xml] → to decommission datanode

```
<property>  
  <name>dfs.hosts.exclude</name>  
  <value>/usr/local/hadoop/etc/hadoop/dfs.exclude</value>  
</property>
```

[yarn-site.xml] → to decommission nodemanager

```
<property>  
  <name>yarn.resourcemanager.nodes.exclude-path</name>  
  <value>/usr/local/hadoop/etc/hadoop/yarn.exclude</value>  
</property>
```

```
hdfs dfsadmin -refreshNodes  
yarn rmadmin -refreshNodes
```

# Now you can safely stop the node

# Decommissioning Results

HDP.tlp - hdadmin@192.168.18.125:22 - Bitvise xterm - hdadmin@master: /usr/local/hadoop/etc/ha

```
hdadmin@master:/usr/local/hadoop/etc/hadoop$ clear
hdadmin@master:/usr/local/hadoop/etc/hadoop$ hdfs dfsadmin -report
2024-12-08 15:41:39,592 WARN util.NativeCodeLoader: Unable to load native
plicable
Configured Capacity: 67314954240 (62.69 GB)
Present Capacity: 56634789888 (52.75 GB)
DFS Remaining: 56634757120 (52.75 GB)
DFS Used: 32768 (32 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
  Missing block groups: 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
```

```
Live datanodes (2):
Name: 192.168.18.123:9866 (slave1)
Hostname: slave1
Decommission Status : Normal
Configured Capacity: 67314954240 (62.69 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 7227572224 (6.73 GB)
DFS Remaining: 56634757120 (52.75 GB)
DFS Used%: 0.00%
DFS Remaining%: 84.13%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Dec 08 15:41:39 UTC 2024
Last Block Report: Sun Dec 08 15:21:58 UTC 2024
Num of Blocks: 0
```

```
Name: 192.168.18.124:9866 (slave2)
Hostname: slave2
Decommission Status : Decommissioned
Configured Capacity: 67314954240 (62.69 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 7205314560 (6.71 GB)
DFS Remaining: 56657014784 (52.77 GB)
DFS Used%: 0.00%
DFS Remaining%: 84.17%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Dec 08 15:41:38 UTC 2024
Last Block Report: Sun Dec 08 15:21:56 UTC 2024
Num of Blocks: 0
```

# Recommisioning Node (All NameNode)

# Start the host you want to add

# Remove hosts

> dfs.exclude

> yarn.exclude

or

echo -n > dfs.exclude

echo -n > yarn.exclude

[Single Node Deployment]

hdfs dfsadmin -refreshNodes

yarn rmadmin -refreshNodes

[HA Deployment (All NameNode)]

hdfs dfsadmin -fs hdfs://<Namenode 01>:8020 -refreshNodes

yarn rmadmin -refreshNodes

# Recommissioning Results (Before vs After)

The left one is before recommissioning, on the right is after recommissioning, the IP is different because of using different network

```
Name: 192.168.18.124:9866 (slave2)
Hostname: slave2
Decommission Status : Decommissioned
Configured Capacity: 67314954240 (62.69 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 7205314560 (6.71 GB)
DFS Remaining: 56657014784 (52.77 GB)
DFS Used%: 0.00%
DFS Remaining%: 84.17%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Sun Dec 08 15:41:38 UTC 2024
Last Block Report: Sun Dec 08 15:21:56 UTC 2024
Num of Blocks: 0
```

```
hdadmin@slave1:~$ hdfs dfsadmin -refreshNodes
2024-12-11 05:44:58,310 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Refresh nodes successful for slave1/192.168.0.50:8020
Refresh nodes successful for slave2/192.168.0.51:8020
hdadmin@slave1:~$ hdfs dfsadmin -fs hdfs://slave1:8020 -refreshNodes
2024-12-11 05:45:42,238 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Refresh nodes successful
hdadmin@slave1:~$ hdfs dfsadmin -report
2024-12-11 05:48:47,604 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Configured Capacity: 134629908480 (125.38 GB)
Present Capacity: 113150894080 (105.38 GB)
DFS Remaining: 113150828544 (105.38 GB)
DFS Used: 65536 (64 KB)
DFS Used%: 0.00%
Replicated Blocks:
  Under replicated blocks: 0
  Blocks with corrupt replicas: 0
  Missing blocks: 0
  Missing blocks (with replication factor 1): 0
  Low redundancy blocks with highest priority to recover: 0
  Pending deletion blocks: 0
Erasure Coded Block Groups:
  Low redundancy block groups: 0
  Block groups with corrupt internal blocks: 0
```

```
Name: 192.168.0.51:9866 (slave2)
Hostname: slave2
Decommission Status : Normal
Configured Capacity: 67314954240 (62.69 GB)
DFS Used: 32768 (32 KB)
Non DFS Used: 7272714240 (6.77 GB)
DFS Remaining: 56589615104 (52.70 GB)
DFS Used%: 0.00%
DFS Remaining%: 84.07%
Configured Cache Capacity: 0 (0 B)
Cache Used: 0 (0 B)
Cache Remaining: 0 (0 B)
Cache Used%: 100.00%
Cache Remaining%: 0.00%
Xceivers: 0
Last contact: Wed Dec 11 05:48:47 UTC 2024
Last Block Report: Wed Dec 11 03:16:28 UTC 2024
Num of Blocks: 0

hdadmin@slave1:~$
```

# Add Nodes

```
rsync -avz <master_node_IP>:/opt/hadoop/etc/hadoop/ /local/directory/  
rsync -avz /path/to/local/file.txt username@remote_host:/path/to/remote/destination/
```

Or you can use **scp** as well

# YARN High Availability

[yarn-site.xml]

```
<property>
  <name>yarn.resourcemanager.ha.enabled</name>
  <value>true</value>
</property>

<property>
  <name>yarn.resourcemanager.cluster-id</name>
  <value>yarn-cluster</value>
</property>

<property>
  <name>yarn.resourcemanager.ha.rm-ids</name>
  <value>rm1,rm2</value>
</property>

<property>
  <name>yarn.resourcemanager.hostname.rm1</name>
  <value>master</value>
</property>

<property>
  <name>yarn.resourcemanager.hostname.rm2</name>
  <value>slave1</value>
</property>

<property>
  <name>yarn.resourcemanager.zk-address</name>
  <value>master:2181,slave1:2181,slave2:2181</value>
</property>
```

```
<property>
  <name>yarn.resourcemanager.recovery.enabled</name>
  <value>true</value>
</property>

<!-- If "yarn.resourcemanager.recovery.enabled" is true -->
<property>
  <name>yarn.resourcemanager.store.class</name>

<value>org.apache.hadoop.yarn.server.resourcemanager.recovery.ZKRMStateStore</value>
</property>
<property>
  <name>yarn.client.failover-proxy-provider</name>
  <value>org.apache.hadoop.yarn.client.ConfiguredRMFailoverProxyProvider</value>
</property>

<!-- Automatic RM Failover -->
<property>
  <name>yarn.resourcemanager.ha.automatic-failover.zk-base-path</name>
  <value>/yarn-leader-election</value>
  <description>Optional setting. The default value is /yarn-leader-
election</description>
</property>
```

# Resource Manager High Availability

```
hdadmin@master:/usr/local/hadoop/etc/hadoop$ yarn rmadmin -transitionToActive rm2
Automatic failover is enabled for org.apache.hadoop.yarn.client.RMHAServiceTarget@3c41ed1d
Refusing to manually manage HA state, since it may cause
a split-brain scenario or other incorrect state.
If you are very sure you know what you are doing, please
specify the --forcemanual flag.
hdadmin@master:/usr/local/hadoop/etc/hadoop$
```

```
HDP.tlp - hdadmin@192.168.18.125:2...
Stopping ZK Failover Controllers on NN hosts [slave1 slave2]
Stopping nodemanagers
Stopping resourcemanager on [ master slave1]
hdadmin@master:/usr/local/hadoop/etc/hadoop$ jps
993 QuorumPeerMain
4819 Jps
2203 ApplicationHistoryServer
hdadmin@master:/usr/local/hadoop/etc/hadoop$ ./sbin/start-all.sh
WARNING: Attempting to start all Apache Hadoop
daemons as hdadmin in 10 seconds.
WARNING: This is not a recommended production d
ployment configuration.
WARNING: Use CTRL-C to abort.
Starting namenodes on [slave1 slave2]
Starting datanodes
Starting journal nodes [slave2 slave1 master]
2024-12-08 15:21:37,543 WARN util.NativeCodeLoa
der: Unable to load native-hadoop library for y
our platform... using builtin-java classes wher
e applicable
Starting ZK Failover Controllers on NN hosts [s
lave1 slave2]
Starting resourcemanager on [ master slave1]
Starting nodemanagers
hdadmin@master:/usr/local/hadoop/etc/hadoop$ jps
993 QuorumPeerMain
5158 JournalNode
5574 ResourceManager
2203 ApplicationHistoryServer
5900 Jps
hdadmin@master:/usr/local/hadoop/etc/hadoop$

HDP.tlp - hdadmin@192.168.18.124:2...
hdadmin@slave2:/usr/local/hadoop/etc/hadoop$ jps
4179 Jps
1039 QuorumPeerMain
hdadmin@slave2:/usr/local/hadoop/etc/hadoop$ jps
4640 DFSZKFailoverController
4517 JournalNode
4391 DataNode
4284 NameNode
5036 Jps
1039 QuorumPeerMain
4895 NodeManager
hdadmin@slave2:/usr/local/hadoop/etc/hadoop$

HDP.tlp - hdadmin@192.168.18.123:2...
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ ya
rn rmadmin -refreshNodes
2024-12-08 15:12:21,358 WARN util.NativeCodeLoa
der: Unable to load native-hadoop library for y
our platform... using builtin-java classes wher
e applicable
2024-12-08 15:12:21,676 INFO client.DefaultNoHA
RMFailoverProxyProvider: Connecting to Resource
Manager at master/192.168.18.125:8033
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ >
dfs.exclude
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ >
yarn.exclude
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ su
do nano yarn-site.xml
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ su
do nano yarn-site.xml
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ jps
1041 QuorumPeerMain
6922 Jps
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$ jps
7264 JournalNode
1041 QuorumPeerMain
7137 DataNode
7554 ResourceManager
7028 NameNode
7721 NodeManager
7867 Jps
7387 DFSZKFailoverController
hdadmin@slave1:/usr/local/hadoop/etc/hadoop$
```

```
HDP.tlp - hdadmin@192.168.18.125:22 - Bitwise xterm - hdadmin@master:/usr/local/hadoop/etc/hadoop
hdadmin@master:/usr/local/hadoop/etc/hadoop$ yarn rmadmin -getServiceState rm1
2024-12-08 15:25:15,172 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where ap
plicable
active
hdadmin@master:/usr/local/hadoop/etc/hadoop$ yarn rmadmin -getServiceState rm2
2024-12-08 15:25:20,257 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where ap
plicable
standby
hdadmin@master:/usr/local/hadoop/etc/hadoop$
```

# MapReduce Uber Task

[mapred-site.xml]

```
<property>
  <name>mapreduce.job.ubertask.enable</name>
  <value>true</value>
</property>

<property>
  <name>mapreduce.job.ubertask.maxmaps</name>
  <value>1</value>
</property>

<property>
  <name>mapreduce.job.ubertask.maxreduces</name>
  <value>1</value>
</property>
```

**Thank you**

